Validity Evidence for the Spiritual Harm and Abuse Scale (SHAS)

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Chapter 1

About this Project

This is a reproducible research project for validation of the nascent Spiritual Harm and Abuse Scale (SHAS), developed by Dan Koch. It aims to function as an evolving technical report to conceptualize and document various validity studies – and, by extension, validity evidence – for this scale. We hope some of these validity considerations and empirical studies amount to meaningful publishable contributions to the nascent research literature in spiritual abuse and/or the literature in religion, counseling psychology, and/or psychological measurement.

The project is built in R using the R Studio IDE. It is built in bookdown and hosted at https://github.com/jackbhuber/SHAS as a public repository. It is served out over Github Pages as a website for collaboration between Jack and Dan at https://jackhuber.org/SHAS/.

1.1 Research Questions

Validation can be an ongoing agenda for research into the meanings and uses of scores from instruments. Following is an initial set of questions to guide research and scale improvement of the SHAS:

- **Item analysis and refinement** - Do some items have too many response categories?
- **Data reduction** - As written and executed, which items appear to be critical (best performing) and which appear to be redundant or noisy? What might a shorter, streamlined instrument look like?
- **Person fit** - How well does the current scale fit the population of examinees? Are there ceiling or floor effects? To what extent might the existing items under-represent the full range of abuse?
- **Scale invariance** - Do the scales mean the same thing to different groups of examinees? To what extent do these scales (abuse constructs) vary by

age, gender, and/or sexuality?

- **Developmental pathway** - Beyond the current conceptualization of scales composed of redundant items, what might an incremental scale of abuse with diagnostic items look like?

Chapter 2

Background of the Project

2.1 Spiritual Harm and Abuse

This is where Dan can introduce and discuss spiritual harm and abuse – his history with the topic, what it might mean in real world terms, the need for scientific study of the topic, and research to date. At a minimum he can paste in text from his extant writings and/or write new text.

2.2 The Spiritual Harm and Abuse Scale (SHAS)

This might be where Dan presents the case for a clinical measure of spiritual harm and abuse and the work he has done to date to develop the SHAS. This could be a summary of his forthcoming JSSR article.

2.3 Validity Research for the SHAS

This might be where I discuss what it means to conduct validity research on any instrument, and in particular the SHAS.

Chapter 3

Content Validity

In this chapter, we consider the content validity of the Spiritual Harm and Abuse Scale (SHAS). We begin by defining content validity and discussing its importance. Then we document the work we did to map the domain of spiritual abuse for the purpose of developing items for piloting and operational use. We conclude by considering what scope exists for more work in this area that would improve future development of SHAS instrumentation.

3.1 About Content Validity

The focus of content validity is "the relationship between the content of a test and the construct it is intended to measure" [American Educational Research Association et al., 2014]. Do they match? Does the instrument contain content that is not part of the construct? Are there parts of the content unmeasured, such that a high score still underrepresents an examinee's true level of a trait?

Content validity in this case concerns the relationship between the concept of spiritual abuse and the content of the 66 items that were piloted to form the SHAS. What is spiritual abuse? What are its important parts? What counts as evidence of it? Is there more to spiritual abuse than was captured by the 66 items?

3.2 Content Specification

"Test developers often work from a specification of the content domain. The content specification carefully describes the content in detail, often with a classification of areas of content and types of items" [American Educational Research Association et al., 2014]. For the purpose of piloting items, the domain of spiritual abuse included two constructs: external (abusive) events and internal states (of emotional harm).

3.2.1 External Events

One domain of spiritual abuse is external abuse *events*. The focus is on something that happened, or a pattern of recurring behavior on the part of someone in a place of spiritual authority that was experienced by the examinee as abusive. The first 52 items of the SHAS were drawn from this domain of possible abusive events. To each item, an examinee was offered five response categories:

1. Never
2. Once or twice
3. Sometimes
4. Often
5. Always

To sum each examinee's vector of responses to these items is to accumulate evidence of abuse. In the aggregate, scores for this collection of items could range from 52 to 260. An examinee who *never* experienced *any* of these events would score 52. At the other extreme, an examinee who scored 260 had *always* experienced *all* of these events.

3.2.2 Internal States

Another important domain of spiritual abuse is *internal states*. The focus is on the subjective experience of an examinee "as a result of negative experiences in church or Christian group settings." The questionnaire included 14 items asking examinees about their emotions in response to events. To each item, an examinee was offered the same five response categories as before:

1. Never
2. Once or twice
3. Sometimes
4. Often
5. Always

To sum each examinee's vector of responses to these items is to accumulate evidence of internal emotional *harm*. Total scores for these items could range from 14 to 70. Examinees who *never* experienced *any* of these unpleasant emotions would score 14, while examinees who scored 70 had *always* experienced *all* of the difficult emotions.

3.2.3 Additional Domains

Reise and Waller [2009] make some interesting points about the nature of the traits addressed in clinical settings which provoke thought about the trait of spiritual abuse.

3.3 Summary

What counts as spiritual abuse? What counts as evidence of it?

Chapter 4

Data Collection

The data were collected online from January 28, 2021 to February 27, 2021.

A final total of 3,222 people responded to the survey at some level. Depending on how much data are missing, this is an impressively large sample size which will provide lots of power for most simple statistics and should for the most part satisfy the large-sample requirements of the advanced statistical procedures associated with measurement work (i.e., factor analysis, IRT).

However, the data come from a convenience sample which raises some challenge of sampling bias. Dan sent the link out to [lots of people], and (if memory serves) encouraged them to send it to people *they* knew. This would seem to bias responses toward people demographically similar to Dan: white, Millennial, west coast American, middle class, highly educated, primarily Protestant and Catholic Christian, etc. This will be more apparent upon closer inspection of the demographics of the sample.

One source of examinees was people who listen to progressive Christians podcasts such as *You Have Permission* which Dan produced at the time of data collection. This is a known source of bias toward people who have some interest in Christianity along with at least some level of rejection of conservative/orthodox Christianity and/or political conservatism. However, this would be an appropriate group from which to invite participation if the target population is people who have experienced spiritual abuse.

Chapter 5

Data Preparation

This chapter documents our decisions surrounding how to prepare the data for analysis. These decisions include details on loading the necessary R packages, loading the data in R, coding or recoding of variables. Also discussed is how we deal with missing data.

5.1 Preliminary work

Load R packages. Some R users load all their packages up front. Others, perhaps for instructional reasons, load them just before calling functions, perhaps for instructional reasons. I can see merit in listing all the packages for properly citing them. Here is a code chunk for loading all the packages up front:

```
# LOAD PACKAGES
library(Hmisc)  # for variable and value labels
library(VIM)  # for missing data
library(naniar)  # for missing data
library(corrplot)  # for correlation plot
library(knitr)  # how RMarkdown compiles code into documents
```

Load data. The data are contained in an Excel file, simply named "data.xlsx", in the "data" folder of this project. The important thing is to keep all the data in the same folder within the larger "SHAS Validity" project folder.

```
# LOAD DATA
library(readxl)  # for reading the source Excel file
shas <- read_excel("data/data.xlsx")  # load Excel file
```

Select items for analysis. Which items should we include for analysis? The first 13 columns in the original data file report data from several screening items that were placed at the beginning of the questionnaire. But because they are

constant and not variable, they don't seem to have a place in scale development. Therefore it seems the important columns range from columns 14 to 97. The first set of items measure the **External Events** scale. These 52 items range from columns 14 to 35. The second set of items measure the **Internal States** scale, and these 14 items range from columns 66 to 79. The last set of items capture **demographic** variables, and these range from columns 80 to 97.

```
# SELECT DATASETS FOR ANALYSIS
all <- shas[, 14:97]  # select items, dems from columns 14-79
items <- shas[, 14:79]  # all Likert items
ext_items <- shas[, 14:35]  # External Events items
int_items <- shas[, 66:79]  # Internal States items

# REMOVE SELECTED COLUMNS
all <- subset(all, select = -c(Ext_Avg:Global_Avg))  # remove columns for now

# OTHER POSSIBLE DATASETS
write.csv(all, "data/all.csv", row.names = FALSE)  # save data for later use
write.csv(items, "data/items.csv", row.names = FALSE)
write.csv(ext_items, "data/ext_items.csv", row.names = FALSE)
write.csv(int_items, "data/int_items.csv", row.names = FALSE)
```

Labeling. This is probably the appropriate place to deal with the labeling of variables and their values. These things should be appropriately labeled to their content, wording, etc. **But I have yet to figure this part out so I will come back to it**.

```
library(Hmisc)
var.labels = c(
# VARIABLE LABELS: Items for External Events Subscale
Q7 = "My pastor/leader explicitly claiming to speak on God's behalf",
Q10 = "Being asked to give up personal, vocational, and/or educational goals by a pastor or
Q11 = "Seeing that anyone who disagreed with my church/pastor was consequently portrayed as
Q16 = "My church community abandoning me in a difficult time",
Q17 = "Being treated as "less than" because of my sexual orientation",
Q18 = "Discouragement of critical thinking by my church or pastor",
Q19 = "Behavior being excessively monitored by my pastor or group members",
Q20 = "Seeing specific members being pressured to give money despite well-known personal fi
Q21 = "Being taught that I would be risking Hell if I left my particular church/group",
Q22 = "Being expected to follow my pastor/leader's personal rules/advice around dating, mar
Q23 = "Prayer replacing needed medical interventions",
Q24 = "Being expected to consult my pastor/leader before making non-religious decisions",
Q25 = "Feeling unable to raise questions and issues",
Q26 = "Being pressured to forgive an abuser while the abuse was ongoing",
Q27 = "Being denied opportunities to serve/volunteer because of my gender",
Q28 = "Seeing the leadership or group protecting abusive individuals",
Q29 = "Feeling special when in my pastor/leader's good graces; feeling ignored when I wasn't
```

```
Q30 = "Being deterred from seeking mental health treatment, counseling and/or medication",
Q31 = "Being shunned or ignored by my pastor or church/group",
Q32 = "Feeling unable to express unhappiness",
Q34 = "Seeing Scripture used to justify physical punishment or other types of severe discipl
Q35 = "Being explicitly taught to distrust my emotions",
Q36 = "Mental or physical problems being interpreted as spiritual/moral weakness",
Q37 = "Being made to feel less spiritually advanced/mature than my pastor/leadership",
Q38 = "Having unrealistic demands placed on my moral or religious behavior",
Q39 = "Being made to feel shame over naturally occurring sexual desires (not actions)",
Q40 = "Threatening Divine punishment to keep group members in line",
Q41 = "Being denied opportunities to serve because of my sexual orientation",
Q42 = "Seeing others shamed or shunned by the pastor/leader or group",
Q43 = "Being shamed by my pastor or the group for poor spiritual/moral performance",
Q44 = "Love or acceptance only being offered if spiritual or moral performance was high",
Q45 = "Vivid descriptions of Hell, Satan, or demons being taught to young children that are
Q46 = "Being treated as "less than" because of my skin color",
Q47 = "Medical care being postponed or not given at all for religious reasons",
Q49 = "The leadership or group not only protecting but elevating abusive individuals",
Q50 = "Being treated as "less than" because of my gender",
Q51 = "Experiencing extreme pressure to take on a role of pastor, missionary, or other spiri
Q52 = "Being blamed for harm that I suffered, rather than blaming those who harmed me",
Q53 = "Seeing Scripture used to justify abusive parent-child behavior",
Q54 = "Being explicitly taught to distrust my intuitions",
Q55 = "Hearing cultural references in sermons that were not commonly known to members of my
Q56 = "Being made to feel like I was crazy or weird for having doubts or questions",
Q57 = "Developing mental or physical ailments as a result of stress from conforming to the g
Q58 = "Terror or horror being used to motivate religious decisions",
Q59 = "Seeing the pastor or group blame the victim of abuse for the abuse itself",
Q60 = "Being encouraged to remain in an abusive marriage by religious leaders",
Q61 = "Vivid descriptions of the end of the world being taught to young children that are de
Q62 = "Seeing Scripture used to justify physical violence",
Q63 = "Being shamed by my pastor or the group for raising questions or concerns",
Q64 = "Witnessing women being pressured to stay in unfaithful or abusive marriages",
Q65 = "Being "cut off" or shunned by more religious family members",
Q66 = "Seeing others treated as "less than" because of their sexual orientation",

# VARIABLE LABELS: Items for Internal States Subscale
Q68 = "Anxiety attacks triggered by religious stimuli",
Q69 = "A lack of spiritual direction or purpose",
Q70 = "Feeling betrayed by God",
Q71 = "Personally avoiding religious activities or settings to reduce distressing feelings",
Q72 = "Feeling as if God harmed me directly",
Q73 = "Self-hatred or self-loathing",
Q74 = "Sadness over the loss of my faith/religious community",
Q76 = "Feeling that I wasted years of my life in a particular church or set of beliefs",
```

```
Q77 = "Anger upon reflecting on negative religious experiences",
Q78 = "A lack of self-worth",
Q79 = "Distrust of God",
Q80 = "Feeling isolated",
Q81 = "Nightmares about my negative religious experiences",
Q82 = "Having trouble navigating life outside my church/community",

# VARIABLE LABELS for Demographics
Q84 = "Age",
Q85 = "Gender",
Q86 = "Sexual orientation",
Q87 = "Race",
Q88 = "Raised in Christian home",
Q89 = "Denomination/group of most negative experience",
Q90 = "Racial makeup of denomination/group of most negative experience",
Q91 = "Race of leadership of denomination/group of most negative experience",
Q92 = "Gender of leadership of denomination/group of most negative experience",
Q95 = "Involvement in denomination/group of most negative experience",
Q93 = "Current religious identification"
)

label(items) = as.list(var.labels[match(names(items), names(var.labels))])
```

5.2 Missing data

Missing data can be an important problem. The problem has two parts. The first is how many data are missing. Obviously the more data are missing, the larger the barrier this problem poses to analysis. The second is the mechanism of missing data. Are the missing data randomly, or is there a pattern to the missing data - like a problematic item, or a distinct group of examinees who chose not to respond? Figure 5.1 presents two plots from the VIM package for missing data [**?**].

```
# VISUALIZE AND QUANTIFY MISSING DATA
library(VIM)
a <- VIM::aggr(all, plot = FALSE)
plot(a, numbers = TRUE, prop = FALSE)
```

The first plot in Figure 5.1 is a frequency distribution of the number of missing data entries for each item. This plot reveals very few missing data for the first set of External Events items, but nearly 200 missing entries for the second set of Internal States items and demographics. The raises two questions. The first is: Who were those examinees? We know that they responded to the first 52 items, then apparently quit the survey. Why? Was it just fatigue, or is there more to know about them? The second question is what to do with the information

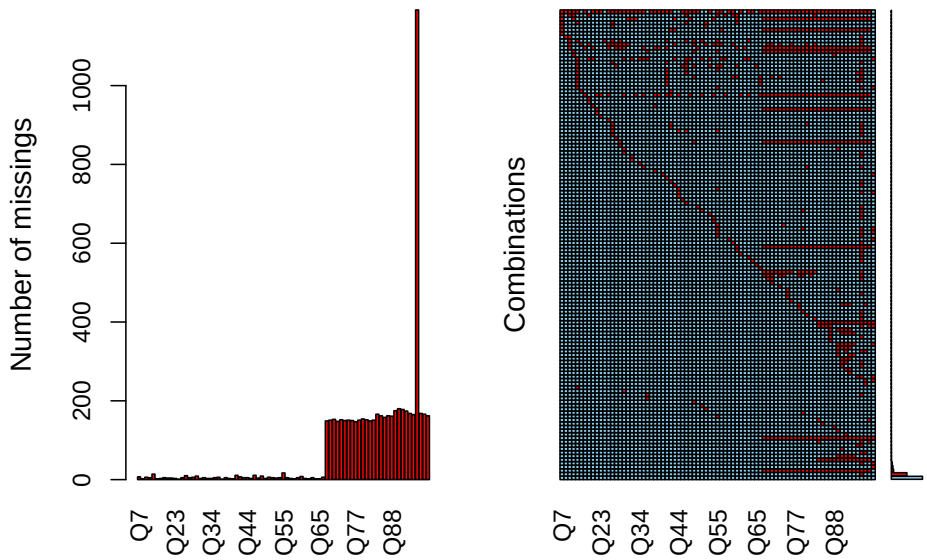


Figure 5.1: Missing Data

they actually did provide.

The second plot in Figure 5.1 illustrates the different combinations of missing entries. The blue squares represent cases with complete responses, and the red indicates missing data. The rows represent combinations of complete and missing responses. The bottom row of entirely blue squares represents complete data – examinees who responded to all of the items. The barplot on the right axis indicates that these were the most frequent combinations. The second most frequent combination was examinees who completed all of the External Events items and then quit the survey. Beyond that, there is no apparent order or pattern to the missing data.

Mechanism of missing data. McKnight et al. [2007] outline a variety of approaches for investigating the mechanisms of missing data. An appropriate first step is to test whether the data are Missing Completely At Random (MCAR) with the use of Little's test Little [1988].

```
library(naniar)
mcar_test(all)
```

```
# A tibble: 1 x 4
  statistic    df p.value missing.patterns
      <dbl> <dbl>   <dbl>            <int>
1     2089.  8663       1              117
```

This test is fairly conclusive. The p-value for the chi-square is 1, which means observed pattern of missing data do not diverge significantly from what would

be expected if the data were missing completely at random.

Were examinees triggered? As mentioned above, a possible explanation of the missing data in the Internal States items is that examinees were triggered. As after responding to the preceding 52 items calling to mind imagery of abusive external events, these examinees were **done** with the survey. A possible test of this hypothesis is an examination of the relationship between a total score of responses to the External Events items and a score of total missingness on the Internal States items. The empirical work to test this hypothesis is laid out in the following code chunk. The items measuring external events are summed into a total score such that a higher value indicates more exposure to abusive external events. The items measuring internal states are coded so that 1=missing and 0=all others response, and these items are summed into a score of total missingness.

```
# 1. MAKE A COPY OF ALL THE ITEMS JUST FOR THIS ANALYSIS
miss_set <- all

# write.csv(miss_set,'data/miss_set.csv') # save as Excel
# to explore visually

# 2. RECODE INTERNAL STATES ITEMS AS DICHOTOMOUS
# (1=MISSING, 0=ENDORSED)
miss_set$Q68[miss_set$Q68 > 0] <- 0
miss_set$Q69[miss_set$Q69 > 0] <- 0
miss_set$Q70[miss_set$Q70 > 0] <- 0
miss_set$Q71[miss_set$Q71 > 0] <- 0
miss_set$Q72[miss_set$Q72 > 0] <- 0
miss_set$Q73[miss_set$Q73 > 0] <- 0
miss_set$Q74[miss_set$Q74 > 0] <- 0
miss_set$Q76[miss_set$Q76 > 0] <- 0
miss_set$Q77[miss_set$Q77 > 0] <- 0
miss_set$Q78[miss_set$Q78 > 0] <- 0
miss_set$Q79[miss_set$Q79 > 0] <- 0
miss_set$Q80[miss_set$Q80 > 0] <- 0
miss_set$Q81[miss_set$Q81 > 0] <- 0
miss_set$Q82[miss_set$Q82 > 0] <- 0
miss_set$Q68[is.na(miss_set$Q68)] <- 1
miss_set$Q69[is.na(miss_set$Q69)] <- 1
miss_set$Q70[is.na(miss_set$Q70)] <- 1
miss_set$Q71[is.na(miss_set$Q71)] <- 1
miss_set$Q72[is.na(miss_set$Q72)] <- 1
miss_set$Q73[is.na(miss_set$Q73)] <- 1
miss_set$Q74[is.na(miss_set$Q74)] <- 1
miss_set$Q76[is.na(miss_set$Q76)] <- 1
miss_set$Q77[is.na(miss_set$Q77)] <- 1
```

```r
miss_set$Q78[is.na(miss_set$Q78)] <- 1
miss_set$Q79[is.na(miss_set$Q79)] <- 1
miss_set$Q80[is.na(miss_set$Q80)] <- 1
miss_set$Q81[is.na(miss_set$Q81)] <- 1
miss_set$Q82[is.na(miss_set$Q82)] <- 1

# 3. OBTAIN TOTAL SCORE ON EXTERNAL EVENTS ITEMS
miss_set$total_ext <- rowSums(miss_set[, c(1:52)])  # total score of exposure to abusive ex
miss_set$total_int <- rowSums(miss_set[, c(53:66)])  # a total score of internal states mis

# 4. RUN CORRELATION BETWEEN TOTAL SCORE AND DICHOTOMOUS
# ITEMS
library(psych)
miss_set2 <- miss_set[, c(53:68)]  # delete all but the internal states and total scores
miss_r <- lowerCor(miss_set2)  # the correlation matrix, rounded to 2 decimals
```

```
    Q68   Q69   Q70   Q71   Q72   Q73   Q74   Q76   Q77   Q78   Q79
Q68  1.00
Q69  0.99  1.00
Q70  0.99  0.99  1.00
Q71  1.00  0.99  0.98  1.00
Q72  0.99  0.99  0.99  0.99  1.00
Q73  1.00  0.99  0.98  0.99  0.99  1.00
Q74  0.99  0.98  0.97  0.99  0.98  0.98  1.00
Q76  0.99  0.99  0.98  0.99  0.99  0.99  0.98  1.00
Q77  0.99  0.99  0.98  1.00  0.98  0.99  0.99  0.99  1.00
Q78  0.99  0.99  0.98  0.99  0.98  0.99  0.98  0.98  0.99  1.00
Q79  0.98  0.98  0.97  0.98  0.97  0.98  0.97  0.97  0.98  0.98  1.00
Q80  0.98  0.98  0.98  0.99  0.98  0.98  0.98  0.99  0.98  0.98  0.97
Q81  1.00  0.99  0.99  1.00  0.99  1.00  0.99  0.99  0.99  0.99  0.98
Q82  0.99  0.99  0.98  0.99  0.98  0.98  0.98  0.99  0.99  0.98  0.97
Q84  0.01  0.00  0.00  0.00  0.03  0.03  0.02  0.02    NA  0.01  0.01
Q85  0.01  0.04  0.03  0.01  0.03  0.01 -0.01  0.04    NA  0.06 -0.01
    Q80   Q81   Q82   Q84   Q85
Q80  1.00
Q81  0.98  1.00
Q82  0.98  0.99  1.00
Q84 -0.02  0.01 -0.01  1.00
Q85  0.05  0.01  0.05  0.06  1.00
```

```r
corPlot(miss_r, stars = TRUE, upper = FALSE)  # examine the many options for this function
```

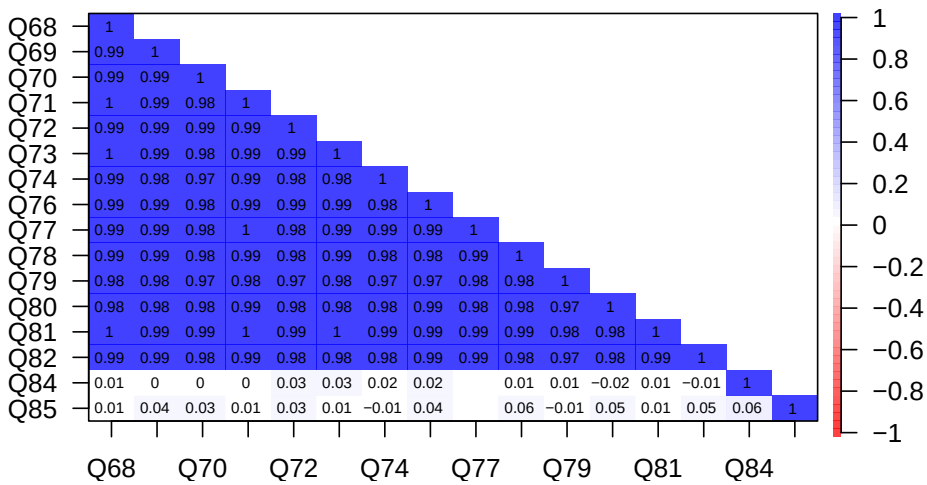


Figure **??** illustrates the correlations among the Internal States items (dichotomized according to their missingness (1=missing, 0=all others)), the total score from the external states (total_ext), and the total score of internal item missingness (total_int). The correlation of total_ext with all the other items is nearly zero, suggesting that more exposure to abusive external states did ***not*** predict nonresponse to the Internal States items. This finding is consistent with the finding from Figure **??** that the missing data appear to missing completely at random.

Missing data methods. There are several different ways to handle missing data. A common method is to *delete all cases missing any data*. This is called **listwise** deletion of data. Its primary advantages are simplicity, expedience, and analysis based on complete information. The disadvantages of listwise are (1) disposal of the information offered by those examinees who responded to some items but not others, and (2) diminishment of the dataset, and with it, a reduction in statistical power. The "all" data file of all Likert items and demographics items derived earlier reports data from 3,222 examinees. The following code chunk creates a subset of complete data. This smaller dataset has 1,841 examinees.

```
# LISTWISE DELETION OF MISSING DATA
all_l <- na.omit(all)  #  Likert items from columns 14-79 plus demographics post listwise
write.csv(all_l, "data/all_l.csv", row.names = FALSE)  # save data

# OTHER POSSIBLE LISTWISE DATASETS

# items <- items[complete.cases(items), ] # create new
# dataset without missing data
items_l <- na.omit(items)  # Likert items from columns 14-79 post listwise
ext_items_l <- na.omit(ext_items)  # Likert items for external events post listwise
int_items_l <- na.omit(int_items)  # Likert items for internal states post listwise
```

```r
write.csv(items_l, "data/items_l.csv", row.names = FALSE)
write.csv(ext_items_l, "data/ext_items_l.csv", row.names = FALSE)
write.csv(int_items_l, "data/int_items_l.csv", row.names = FALSE)
```

Analysis can proceed with these two datasets:

1. the "all" dataset of all Likert and demographic items (N=3,222)
2. the "all_l" dataset of 1,841 examinees who reported complete data

Chapter 6

Item Analysis

In this chapter, we examine the technical quality of the items that Dan piloted for the SHAS. Here we address questions like the following:

- Which items, if any, are too "easy"? What is the impact of this?
- Which items, if any, are too "difficult"? What is the impact of this?
- Which items, if any, correlate weakly with others (and therefore might need to be refined or excluded from scaling?
- Which items, if any, have too many response categories which might be profitably collapsed?

We begin by defining item analysis and discussing its importance. Then we document our item analyses from an item response theory (IRT) framework. We conclude by summarizing what we learn from these analyses to help us refine these 66 items and possibly develop better ones in the future.

6.1 Item analyses in item response theory

Item response theory (IRT) is an established framework for empirically evaluating what items measure [Embretson and Reise, 2000, Wright and Masters, 1982]. IRT methods complement methods derived from classical test theory in several ways.

1. **Examinees and items on the same equal-interval scale**. In classical test theory, examinees differ in their scores and items differ in their means or percent of examinees selecting a response; examinees and items are on different scales. By contrast, IRT methods allow measurement of persons and items on the ***same scale*** with *equal interval properties* of the *logit* scale** and resulting linear measures (Wright & Stone, 1999). In IRT we can estimate *item parameters independently of the characteristics of the*

calibrating sample, and we can estimate *person parameters apart from the difficulties of the items taken* (Masters, 1982; Rasch, 1966).

2. **Item separation has diagnostic value**. A core assumption of CTT is that items are *redundant*; they contribute meaningful variation to a total score but they do not occupy a distinct place on the score continuum. IRT is different. IRT does not assume that items are redundant; instead it investigates whether items make distinct contributions to the measurement of the latent trait.

Whereas classical test theory measurement is principally concerned with the ratio of noise to signal in the variation contributed by an item, IRT models compute the probability of a certain response to an item given the level of the latent construct the individual possesses (i.e., spiritual abuse) and the relevant item's difficulty of endorsement. IRT is based on the logistic model and operates on the *logit* scale.

In what follows, we conduct item analyses through a comparison of several different IRT models: the Partial Credit Model (PCM), the Generalized Partial Credit Model (GPCM), and the Rating Scale Model (RSM). I credit Okan Bulut [**?**] for his excellent tutorial illustrating how to carry out IRT analyses in R using the `mirt` package (cite mirt).

6.1.1 IRT Model Assumptions

IRT makes two assumptions of the data. The first is unidimensionality of the latent trait. Is the trait one dimension? This is important for the SHAS because the questionnaire explicitly distinguished between items measuring external events and internal states. To check this assumption we first confirm the factor structure using confirmatory factor analysis (CFA) and Rasch Principal Component Analysis of Residuals (PCAR; Smith, 2002).

The second assumption is local independence [**?**]. This means that any set of items should not share any meaningful correlation once the latent variable has been accounted for (Edwards, Houts, & Cai, 2018). To check the assumption of local independence, we used the Q3 statistic (Yen, 1984). The Q3 statistic index criteria are that the raw residual correlation between pairs of items should never exceed 0.10 (Marais & Andrich, 2008).

Using these criteria, we found XX items having local dependence. The items that had the highest raw residual correlations had negative signs and no positive residual correlation. In other words, the assumption of local independence in this study was met.

6.1.2 Partial Credit Model (PCM)

The first model we examine is the Partial Credit Model (PCM) [**?**]. This is a polytomous item response model belonging to the Rasch family. This means

that it extends the Rasch model, which is based on dichotomous item responses (1=correct response, 0=incorrect response), by considering the probabliity of choosing from among several response options such as those typically presented in Likert scales of agreement or frequency. The PCM assumes that items use ordered response categories as they exist in questionnaires, such as unidimensional rating scales. The number of response categories in each item may vary [Embretson and Reise, 2000]. It follows that the PCM contains m (m + 1 being the number of response categories) threshold parameters. Each threshold parameter marks a category intersection [**?**]. Threshold sometimes refers to "step difficulty" or "step parameter" and illustrates the point on the latent trait continuum (e.g., spiritual abuse) where a response in category k becomes more likely than a response in category k − 1 (de Ayala, 2009). The PCM provides scores for items, persons, and step parameters on a logit scale.

Model Estimation. We begin by loading the `mirt` library and estimating the model on the data from the 66 SHAS items. This model constrains the slopes to 1 and freely estimates the variance parameters.

```
# ESTIMATE MIRT MODEL
library(mirt)
items <- read.csv("data/items.csv")
model.pcm <- "SPIRITUAL ABUSE = 1-66"
results.pcm <- mirt(data = items, model = model.pcm, itemtype = "Rasch",
    verbose = FALSE)
```

```
coef.pcm <- coef(results.pcm, IRTpars = TRUE, simplify = TRUE)
items.pcm <- as.data.frame(coef.pcm$items)
# print(results.pcm)
knitr::kable(round(items.pcm, 2), caption = "Partial Credit Model (PCM) - Item Parameters")
```

Item parameters. Table 6.1 reports the item parameters estimated by the PCM. The first column shows the discrimination parameter, which is equal to "1" for all items because the Partial Credit Model, similar to the Rasch model, constrains the discrimination parameter to be "1". In the following columns (b1 to b4) are the estimated thresholds or step parameters. Because the SHAS items all have five response categories, the Partial Credit Model estimates four threshold parameters (b1 to b4) for each item. The question here is the spread of the parameters across the range of the trait. In the PCM, "the property of ordered threshold parameters is not a requirement in the partial credit or generalized partial credit models" [Embretson and Reise, 2000].

Consider the first item: Q7. The thresholds are as follows: b1=0.23, b2=0.08, b3=0.74, b4=1.11. The threshold for b1 is higher than b2. This means examinees who experienced more spiritual abuse overall were more likely to choose "Never" than "Once or twice" on this item.

Item fit. Two fit statistics assess the fit of an item to the PCM: the infit mean square (MNSQ) and the outfit mean square (MNSQ). The infit statistic places

Table 6.1: Partial Credit Model (PCM) - Item Parameters

	a	b1	b2	b3	b4
Q7	1	0.23	0.08	0.74	1.11
Q10	1	1.26	0.67	1.47	1.78
Q11	1	-0.05	0.20	0.69	1.53
Q16	1	0.50	0.59	1.17	1.64
Q17	1	3.93	0.39	1.20	1.18
Q18	1	-0.15	-0.39	0.71	1.28
Q19	1	0.42	0.20	0.94	1.69
Q20	1	0.97	0.37	0.99	1.24
Q21	1	1.11	0.38	0.93	0.78
Q22	1	-0.39	-0.61	-0.27	0.38
Q23	1	0.77	0.64	1.62	2.19
Q24	1	1.13	0.47	1.55	1.98
Q25	1	-0.95	-0.81	0.18	1.08
Q26	1	1.41	0.74	1.03	1.55
Q27	1	1.64	0.36	1.09	1.20
Q28	1	0.08	0.51	1.07	1.89
Q29	1	0.12	0.04	0.88	1.58
Q30	1	0.89	0.38	1.23	1.92
Q31	1	0.51	0.64	1.49	1.88
Q32	1	-0.30	-0.53	0.69	1.53
Q34	1	0.10	0.04	0.94	1.57
Q35	1	-0.62	-0.70	0.29	0.82
Q36	1	-0.47	-0.24	0.60	1.43
Q37	1	-0.54	-0.44	0.49	1.06
Q38	1	-0.43	-0.49	0.52	0.92
Q39	1	-0.16	-0.62	0.29	0.65
Q40	1	0.66	0.21	1.09	1.17
Q41	1	3.89	0.65	1.48	1.10
Q42	1	-0.59	0.03	0.95	1.75
Q43	1	0.50	0.57	1.46	2.04
Q44	1	-0.01	-0.17	0.70	1.57
Q45	1	-0.10	-0.12	0.57	0.92
Q46	1	3.92	1.19	2.64	2.40
Q47	1	1.56	1.29	2.47	2.21
Q49	1	0.42	0.57	1.22	1.90
Q50	1	1.58	-0.24	0.78	1.15
Q51	1	0.44	0.33	0.82	1.42
Q52	1	0.73	0.54	1.06	1.93
Q53	1	0.36	0.17	1.00	1.43
Q54	1	-0.36	-0.48	0.54	0.97
Q55	1	1.34	1.18	2.00	2.39
Q56	1	-0.18	-0.24	0.56	1.47
Q57	1	0.83	0.33	1.04	1.56
Q58	1	0.04	-0.13	0.97	1.45
Q59	1	0.38	0.42	1.21	2.07
Q60	1	2.21	0.95	1.46	2.19
Q61	1	0.05	-0.24	0.49	0.93
Q62	1	0.54	0.20	1.38	1.96
Q63	1	0.05	0.33	1.19	1.85
Q64	1	0.19	0.45	1.18	1.63
Q65	1	1.22	0.78	1.39	1.82

greater emphasis on unexpected responses that are close to the examinees and item location. The outfit is sensitive to unexpected responses that are far from the location [Bond and Fox, 2015]. The infit and outfit values identify potential unexpected response patterns. The expected value of infit or outfit for each item is 1.0, with a range of acceptable values ranging from 0.5 to 1.5. Values outside this boundary indicate a lack of fit between items and models.

```
itemfit(results.pcm, na.rm = TRUE, "infit")
```

```
   item outfit z.outfit infit z.infit
1    Q7  1.276    7.187 1.122   4.669
2   Q10  0.963   -0.671 0.988  -0.415
3   Q11  0.780   -7.399 0.799  -8.713
4   Q16  0.930   -1.824 0.939  -2.337
5   Q17  2.813    7.053 1.555   9.545
6   Q18  0.777   -7.906 0.804  -8.498
7   Q19  0.778   -6.464 0.836  -6.867
8   Q20  0.970   -0.618 1.014   0.513
9   Q21  1.203    3.704 1.043   1.575
10  Q22  1.104    2.990 1.099   3.596
11  Q23  0.941   -1.383 1.007   0.269
12  Q24  0.913   -1.766 0.936  -2.327
13  Q25  0.771   -9.221 0.791  -9.059
14  Q26  1.313    4.798 0.963  -1.247
15  Q27  2.671   17.851 1.502  15.055
16  Q28  0.848   -4.823 0.869  -5.376
17  Q29  1.014    0.428 1.026   1.060
18  Q30  0.848   -3.539 0.893  -4.155
19  Q31  1.012    0.318 1.006   0.216
20  Q32  0.882   -4.193 0.897  -4.308
21  Q34  1.119    3.474 1.102   3.963
22  Q35  0.976   -0.850 0.917  -3.397
23  Q36  0.792   -7.866 0.812  -8.212
24  Q37  0.845   -5.731 0.862  -5.881
25  Q38  0.680  -12.407 0.711 -13.043
26  Q39  0.994   -0.167 0.998  -0.085
27  Q40  0.788   -5.520 0.845  -6.321
28  Q41  6.705   15.057 1.534   8.204
29  Q42  0.763   -9.409 0.790  -9.222
30  Q43  0.748   -7.211 0.795  -8.281
31  Q44  0.715   -9.971 0.734 -11.859
32  Q45  0.978   -0.672 0.994  -0.251
33  Q46  3.223    8.213 1.392   4.076
34  Q47  0.986   -0.207 1.027   0.728
35  Q49  0.821   -5.079 0.865  -5.373
36  Q50  1.947   12.371 1.513  16.610
```

37	Q51	1.287	6.901	1.232	8.445
38	Q52	0.698	-7.911	0.759	-9.911
39	Q53	0.836	-4.767	0.896	-4.247
40	Q54	0.872	-4.519	0.866	-5.674
41	Q55	2.828	22.030	1.577	14.747
42	Q56	0.659	-12.674	0.687	-14.270
43	Q57	0.800	-4.843	0.896	-4.083
44	Q58	0.879	-3.921	0.920	-3.294
45	Q59	0.720	-8.510	0.778	-9.340
46	Q60	1.436	4.429	1.199	4.970
47	Q61	1.013	0.402	1.025	0.997
48	Q62	0.968	-0.821	1.037	1.426
49	Q63	0.722	-9.474	0.775	-9.632
50	Q64	0.900	-2.977	0.913	-3.471
51	Q65	1.183	3.247	1.051	1.715
52	Q66	1.013	0.443	0.984	-0.613
53	Q68	0.890	-2.575	0.959	-1.542
54	Q69	1.184	6.976	1.185	7.178
55	Q70	1.238	6.218	1.223	7.897
56	Q71	0.869	-4.950	0.891	-4.589
57	Q72	1.020	0.380	1.091	2.772
58	Q73	1.089	2.783	1.121	4.679
59	Q74	1.205	6.924	1.188	7.246
60	Q76	0.922	-2.415	0.925	-3.068
61	Q77	0.735	-11.202	0.745	-11.386
62	Q78	1.024	0.878	1.039	1.571
63	Q79	1.215	6.432	1.184	6.889
64	Q80	0.938	-2.471	0.946	-2.227
65	Q81	0.812	-4.346	0.909	-3.187
66	Q82	1.069	2.012	1.127	4.750

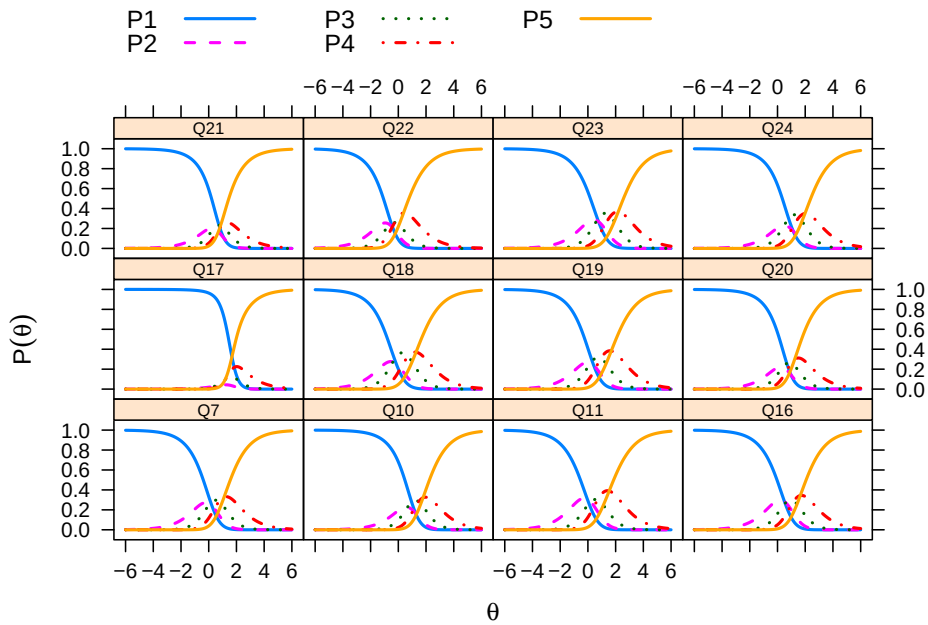
As seen in the **??**tab:pcm-item-fit), XX items had infit and outfit statistics within an acceptable value (0.5 - 1.5). XX misfitting items were found and this means that XX items fit within the Rasch PCM. [1]

Item plots. We can use also examine these item statistics visually. The following curves are called category response curves because they represent "the probability of an examinee (or survey respondent) responding in a particular category conditional on trait level" [Embretson and Reise, 2000]. In general, "the higher the slope parameters, the steeper the operating characteristic curves and the more narrow and peaked the category response curves, indicating that the response categories differentiate among trait levels fairly well" [Embretson and Reise, 2000]. Polytomous IRT models such as the Partial Credit Model have option characteristic curves (OCCs) which can be considered as an extension of item characteristic curves (ICCs) for polytomous items. Because the items have

[1] and this is important because…

more than two response categories, OCCs have multiple curves in the same plot. Each curve represents the probability of selecting a particular response option as a function of the latent trait.

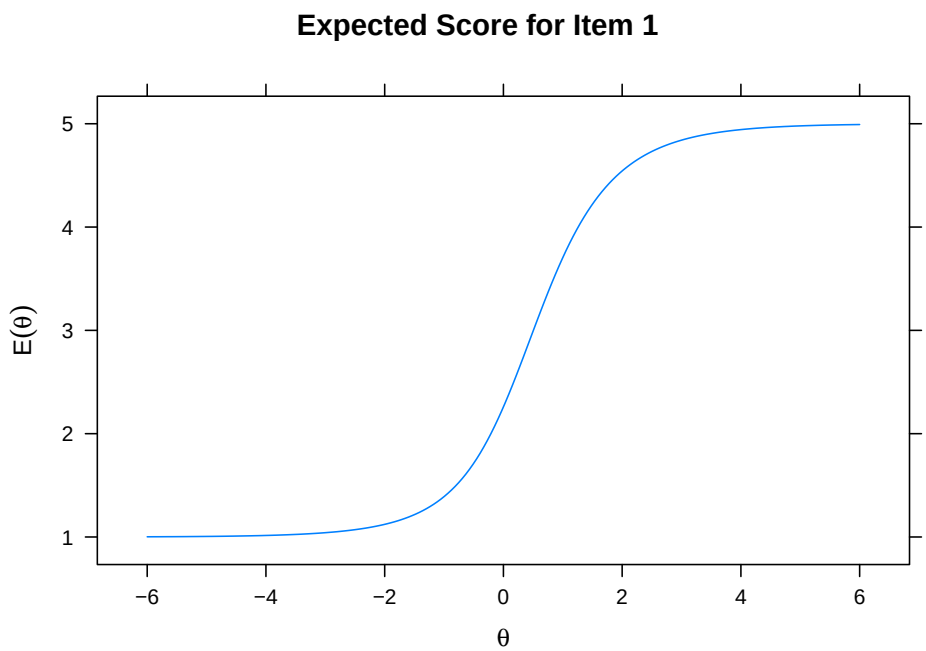
```
# PCM ITEM PLOTS itemplot(results.pcm, 1, type='infotrace')
plot(results.pcm, type = "trace", which.items = c(1, 2, 3, 4,
    5, 6, 7, 8, 9, 10, 11, 12), main = "", par.settings = simpleTheme(lty = 1:4,
    lwd = 2), auto.key = list(points = FALSE, lines = TRUE, columns = 4))  # option curves
```



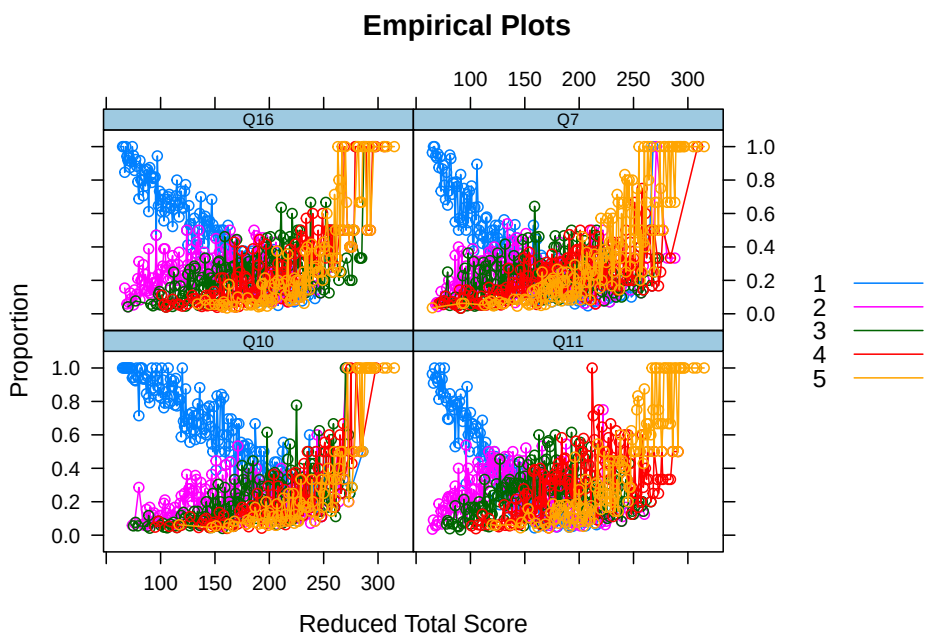
In Figure ??fig:pcm-occs), each item has four OCCs representing the four response options. The response categories are labeled as "P1" to "P4". For all four items, the OCCs follow the same order as the response categories. The OCC for the first response option (P1) is on the very left side of the plot, whereas the last response option (P4) is located on the right side of the plot. There are two questions to ask of OCCs: (1) Are they in the correct order? and (2) Does each response category meaningfully represent the responses of respondents at a given trait level? In this case, the OCCs are in the correct order, but the middle categories do not seem to capture distinct respondents.

Model fit. One way to evaluate the fit of the Partial Credit Model is to evaluate how well it fits the data from the item responses. The following table presents this information.

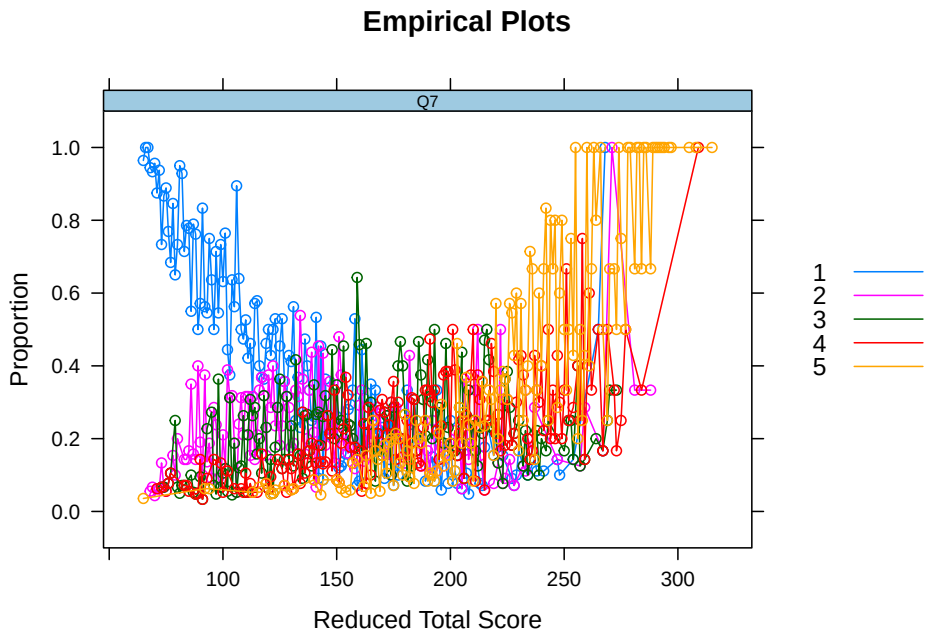
```
# PCM MODEL FIT
itemplot(results.pcm, 1, type = "score")
```




```
empirical_plot(items, c(1, 2, 3, 4), main = "Empirical Plots")  # multiple items
```




```
empirical_plot(items, 1, main = "Empirical Plots")  # one item
```



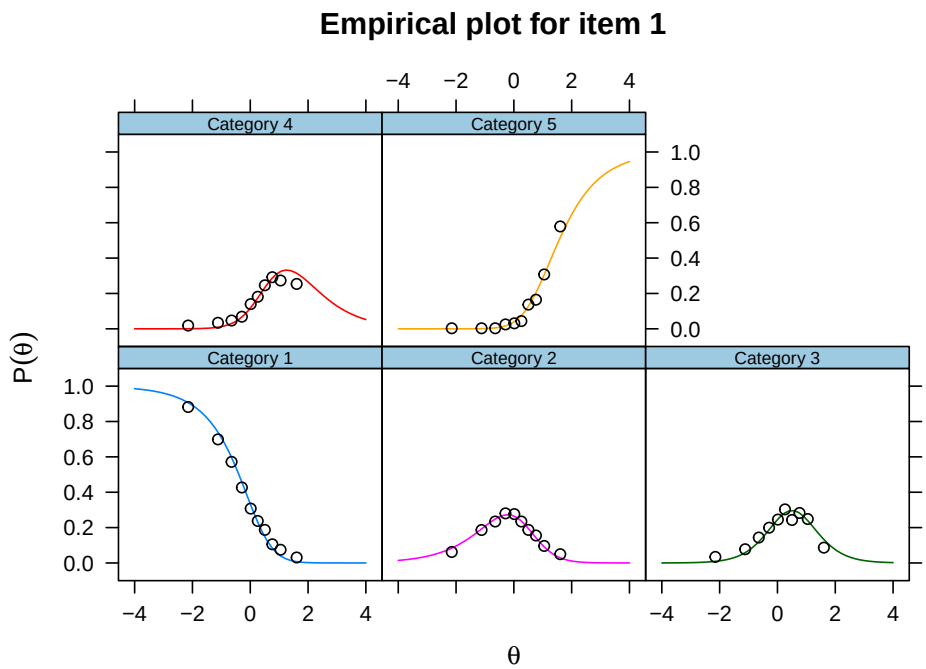

```
pcmfit <- as.data.frame(itemfit(results.pcm, na.rm = TRUE))
print(pcmfit, caption = "Partial Credit Model (PCM) - Item Fit Statistics")
```

```
   item      S_X2 df.S_X2   RMSEA.S_X2       p.S_X2
1    Q7  533.3564     533 0.0004738806 4.875015e-01
2   Q10  495.1835     467 0.0045017043 1.772288e-01
3   Q11  579.6394     531 0.0055460592 7.083978e-02
4   Q16  481.6413     503 0.0000000000 7.460408e-01
5   Q17  443.3413     213 0.0190560901 2.710678e-18
6   Q18  511.1011     523 0.0000000000 6.368612e-01
7   Q19  548.8232     517 0.0045463620 1.609217e-01
8   Q20  543.2461     513 0.0044495252 1.718095e-01
9   Q21  556.8474     518 0.0050182679 1.153414e-01
10  Q22  610.6849     477 0.0097010700 3.161075e-05
11  Q23  451.2542     440 0.0029306759 3.450871e-01
12  Q24  439.9162     449 0.0000000000 6.113781e-01
13  Q25  505.6217     492 0.0030490903 3.258128e-01
14  Q26  481.9682     475 0.0022194765 4.026140e-01
15  Q27  896.8917     479 0.0171159793 9.180280e-28
16  Q28  514.6797     500 0.0031398698 3.153365e-01
17  Q29  530.1984     524 0.0019930273 4.163283e-01
18  Q30  569.1584     488 0.0074729914 6.434023e-03
19  Q31  526.1497     479 0.0057492248 6.714310e-02
20  Q32  486.5294     515 0.0000000000 8.114182e-01
21  Q34  586.1511     525 0.0062540338 3.296610e-02
```

22	Q35	542.2495	508	0.0047580948	1.418454e-01
23	Q36	553.8355	515	0.0050320936	1.147646e-01
24	Q37	548.0106	519	0.0043324357	1.827871e-01
25	Q38	568.9640	517	0.0058095664	5.643818e-02
26	Q39	473.9639	507	0.0000000000	8.507912e-01
27	Q40	573.7718	529	0.0053310428	8.694811e-02
28	Q41	435.2054	180	0.0218195764	3.605120e-23
29	Q42	549.9676	494	0.0061679751	4.106454e-02
30	Q43	502.6761	461	0.0055097377	8.770565e-02
31	Q44	618.0733	528	0.0075686550	4.042814e-03
32	Q45	548.4064	531	0.0033177605	2.915691e-01
33	Q46	249.0751	108	0.0209435812	3.719748e-13
34	Q47	347.5054	338	0.0030730106	3.490803e-01
35	Q49	515.8695	493	0.0039467760	2.301057e-01
36	Q50	1193.8223	508	0.0212917621	3.932729e-57
37	Q51	613.6675	527	0.0074312269	5.292971e-03
38	Q52	486.9722	493	0.0000000000	5.680840e-01
39	Q53	541.7427	523	0.0034689851	2.764940e-01
40	Q54	561.7842	518	0.0053275956	8.941954e-02
41	Q55	977.8732	373	0.0233354077	9.345000e-56
42	Q56	700.4023	524	0.0106322243	3.610390e-07
43	Q57	508.4694	507	0.0009865056	4.732781e-01
44	Q58	583.6834	531	0.0057720141	5.632886e-02
45	Q59	546.5812	477	0.0069988128	1.488867e-02
46	Q60	552.8959	347	0.0141155160	1.226614e-11
47	Q61	559.8063	528	0.0044975646	1.634943e-01
48	Q62	511.0822	489	0.0038940741	2.366890e-01
49	Q63	557.6752	505	0.0059182722	5.219718e-02
50	Q64	560.6293	508	0.0058981997	5.282423e-02
51	Q65	493.0967	466	0.0044187815	1.860160e-01
52	Q66	484.6288	486	0.0000000000	5.090283e-01
53	Q68	524.0797	483	0.0053441398	9.555432e-02
54	Q69	514.9555	493	0.0038671045	2.387767e-01
55	Q70	596.9121	469	0.0095698906	5.464063e-05
56	Q71	517.4845	517	0.0005609969	4.857223e-01
57	Q72	398.9107	382	0.0038555564	2.653161e-01
58	Q73	575.2774	521	0.0059146412	4.979387e-02
59	Q74	656.6494	501	0.0102139223	3.324529e-06
60	Q76	578.0990	526	0.0057671246	5.745455e-02
61	Q77	556.0596	489	0.0067859959	1.901382e-02
62	Q78	517.0659	518	0.0000000000	5.033200e-01
63	Q79	565.0319	517	0.0055854376	7.072720e-02
64	Q80	495.3246	488	0.0022450114	3.995138e-01
65	Q81	431.9793	407	0.0045397375	1.890289e-01
66	Q82	520.8786	480	0.0053476755	9.594222e-02

```r
knitr::kable(pcmfit, caption = "Partial Credit Model (PCM) - Item Fit Statistics")
```

```r
itemfit(results.pcm, empirical.plot = 1)
```




```r
itemfit(results.pcm, "infit", na.rm = TRUE)
```

	item	outfit	z.outfit	infit	z.infit
1	Q7	1.276	7.187	1.122	4.669
2	Q10	0.963	-0.671	0.988	-0.415
3	Q11	0.780	-7.399	0.799	-8.713
4	Q16	0.930	-1.824	0.939	-2.337
5	Q17	2.813	7.053	1.555	9.545
6	Q18	0.777	-7.906	0.804	-8.498
7	Q19	0.778	-6.464	0.836	-6.867
8	Q20	0.970	-0.618	1.014	0.513
9	Q21	1.203	3.704	1.043	1.575
10	Q22	1.104	2.990	1.099	3.596
11	Q23	0.941	-1.383	1.007	0.269
12	Q24	0.913	-1.766	0.936	-2.327
13	Q25	0.771	-9.221	0.791	-9.059
14	Q26	1.313	4.798	0.963	-1.247
15	Q27	2.671	17.851	1.502	15.055
16	Q28	0.848	-4.823	0.869	-5.376
17	Q29	1.014	0.428	1.026	1.060
18	Q30	0.848	-3.539	0.893	-4.155

Table 6.2: Partial Credit Model (PCM) - Item Fit Statistics

item	S_X2	df.S_X2	RMSEA.S_X2	p.S_X2
Q7	533.3564	533	0.0004739	0.4875015
Q10	495.1835	467	0.0045017	0.1772288
Q11	579.6394	531	0.0055461	0.0708398
Q16	481.6413	503	0.0000000	0.7460408
Q17	443.3413	213	0.0190561	0.0000000
Q18	511.1011	523	0.0000000	0.6368612
Q19	548.8232	517	0.0045464	0.1609217
Q20	543.2461	513	0.0044495	0.1718095
Q21	556.8474	518	0.0050183	0.1153414
Q22	610.6849	477	0.0097011	0.0000316
Q23	451.2542	440	0.0029307	0.3450871
Q24	439.9162	449	0.0000000	0.6113781
Q25	505.6217	492	0.0030491	0.3258128
Q26	481.9682	475	0.0022195	0.4026140
Q27	896.8917	479	0.0171160	0.0000000
Q28	514.6797	500	0.0031399	0.3153365
Q29	530.1984	524	0.0019930	0.4163283
Q30	569.1584	488	0.0074730	0.0064340
Q31	526.1497	479	0.0057492	0.0671431
Q32	486.5294	515	0.0000000	0.8114182
Q34	586.1511	525	0.0062540	0.0329661
Q35	542.2495	508	0.0047581	0.1418454
Q36	553.8355	515	0.0050321	0.1147646
Q37	548.0106	519	0.0043324	0.1827871
Q38	568.9640	517	0.0058096	0.0564382
Q39	473.9639	507	0.0000000	0.8507912
Q40	573.7718	529	0.0053310	0.0869481
Q41	435.2054	180	0.0218196	0.0000000
Q42	549.9676	494	0.0061680	0.0410645
Q43	502.6761	461	0.0055097	0.0877056
Q44	618.0733	528	0.0075687	0.0040428
Q45	548.4064	531	0.0033178	0.2915691
Q46	249.0751	108	0.0209436	0.0000000
Q47	347.5054	338	0.0030730	0.3490803
Q49	515.8695	493	0.0039468	0.2301057
Q50	1193.8223	508	0.0212918	0.0000000
Q51	613.6675	527	0.0074312	0.0052930
Q52	486.9722	493	0.0000000	0.5680840
Q53	541.7427	523	0.0034690	0.2764940
Q54	561.7842	518	0.0053276	0.0894195
Q55	977.8732	373	0.0233354	0.0000000
Q56	700.4023	524	0.0106322	0.0000004
Q57	508.4694	507	0.0009865	0.4732781
Q58	583.6834	531	0.0057720	0.0563289
Q59	546.5812	477	0.0069988	0.0148887
Q60	552.8959	347	0.0141155	0.0000000
Q61	559.8063	528	0.0044976	0.1634943
Q62	511.0822	489	0.0038941	0.2366890
Q63	557.6752	505	0.0059183	0.0521972
Q64	560.6293	508	0.0058982	0.0528242
Q65	493.0967	466	0.0044188	0.1860160

19	Q31	1.012	0.318	1.006	0.216
20	Q32	0.882	-4.193	0.897	-4.308
21	Q34	1.119	3.474	1.102	3.963
22	Q35	0.976	-0.850	0.917	-3.397
23	Q36	0.792	-7.866	0.812	-8.212
24	Q37	0.845	-5.731	0.862	-5.881
25	Q38	0.680	-12.407	0.711	-13.043
26	Q39	0.994	-0.167	0.998	-0.085
27	Q40	0.788	-5.520	0.845	-6.321
28	Q41	6.705	15.057	1.534	8.204
29	Q42	0.763	-9.409	0.790	-9.222
30	Q43	0.748	-7.211	0.795	-8.281
31	Q44	0.715	-9.971	0.734	-11.859
32	Q45	0.978	-0.672	0.994	-0.251
33	Q46	3.223	8.213	1.392	4.076
34	Q47	0.986	-0.207	1.027	0.728
35	Q49	0.821	-5.079	0.865	-5.373
36	Q50	1.947	12.371	1.513	16.610
37	Q51	1.287	6.901	1.232	8.445
38	Q52	0.698	-7.911	0.759	-9.911
39	Q53	0.836	-4.767	0.896	-4.247
40	Q54	0.872	-4.519	0.866	-5.674
41	Q55	2.828	22.030	1.577	14.747
42	Q56	0.659	-12.674	0.687	-14.270
43	Q57	0.800	-4.843	0.896	-4.083
44	Q58	0.879	-3.921	0.920	-3.294
45	Q59	0.720	-8.510	0.778	-9.340
46	Q60	1.436	4.429	1.199	4.970
47	Q61	1.013	0.402	1.025	0.997
48	Q62	0.968	-0.821	1.037	1.426
49	Q63	0.722	-9.474	0.775	-9.632
50	Q64	0.900	-2.977	0.913	-3.471
51	Q65	1.183	3.247	1.051	1.715
52	Q66	1.013	0.443	0.984	-0.613
53	Q68	0.890	-2.575	0.959	-1.542
54	Q69	1.184	6.976	1.185	7.178
55	Q70	1.238	6.218	1.223	7.897
56	Q71	0.869	-4.950	0.891	-4.589
57	Q72	1.020	0.380	1.091	2.772
58	Q73	1.089	2.783	1.121	4.679
59	Q74	1.205	6.924	1.188	7.246
60	Q76	0.922	-2.415	0.925	-3.068
61	Q77	0.735	-11.202	0.745	-11.386
62	Q78	1.024	0.878	1.039	1.571
63	Q79	1.215	6.432	1.184	6.889
64	Q80	0.938	-2.471	0.946	-2.227

```
65  Q81  0.812   -4.346 0.909  -3.187
66  Q82  1.069    2.012 1.127   4.750
```

Table X "contains estimation results and the fit index model from each item of the SHAS." The estimates of item difficulty were between -X.XX TO X.XX (ie., -1.36 to 1.51) on the logit scale. Based on the item discrimination index (PTBIS; point-biserial correlation), which is analogous to classical test theory (item to-total-correlation), X SHAS items have high discrimination values in a positive direction, with no values below 0.30 or negative values. This indicates that XX items function well to distinguish persons with high versus low levels of spiritual abuse. For each item, we examined the step parameter of category endorsements, and the statistics of each response for the items. X items, except item X, followed the step ordering requirement. We found Item X (Step 1 = -0.XX, Step 2 = -0.XX, Step 3 = 0.XX) experienced threshold disordering as the thresholds were not ordered from lowest to highest. However, none of the infit and outfit statistics for response categories were greater than 2. This finding indicated that collapsing categories was not necessary because the disordered threshold still fit and did not violate the Rasch model (Linacre, 2010; 2018). Based on this information, we concluded that, in general, the SHAS items (fit)/(did not fit) the Rasch PCM model.

6.1.3 Generalized Partial Credit Model (GPCM)

In the PCM, "the property of ordered threshold parameters is not a requirement in the partial credit or generalized partial credit models" [Embretson and Reise, 2000].

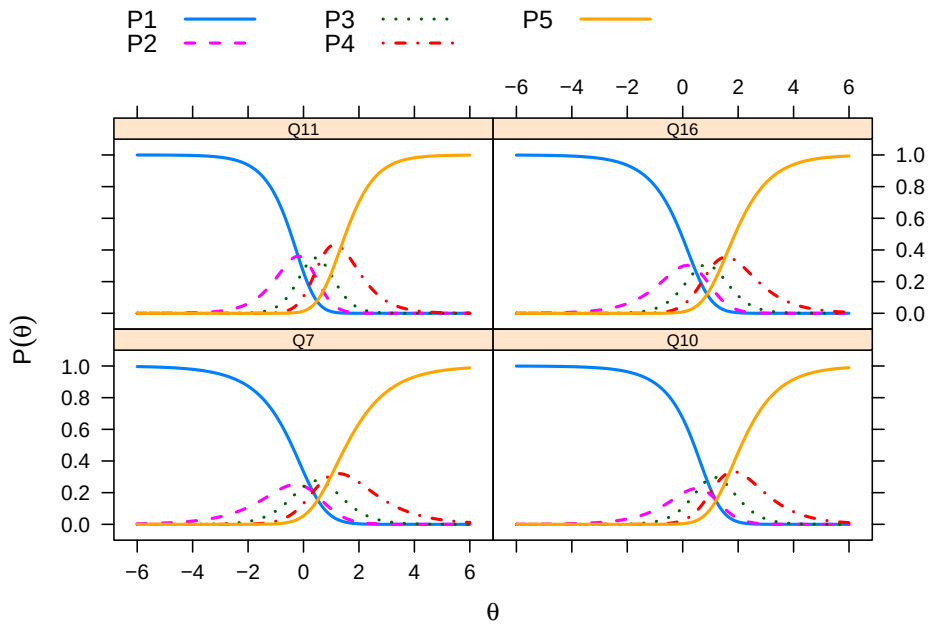
```r
# GENERALIZED PCM ESTIMATION
model.gpcm <- "SPIRITUAL ABUSE GPCM = 1-66"
results.gpcm <- mirt(data = items, model = model.gpcm, itemtype = "gpcm",
    verbose = FALSE)
coef.gpcm <- coef(results.gpcm, IRTpars = TRUE, simplify = TRUE)
items.gpcm <- as.data.frame(coef.gpcm$items)
print(results.gpcm)
```

```
Call:
mirt(data = items, model = model.gpcm, itemtype = "gpcm", verbose = FALSE)

Full-information item factor analysis with 1 factor(s).
Converged within 1e-04 tolerance after 56 EM iterations.
mirt version: 1.36.1
M-step optimizer: BFGS
EM acceleration: Ramsay
Number of rectangular quadrature: 61
Latent density type: Gaussian

Log-likelihood = -234269.7
```

```
Estimated parameters: 330
AIC = 469199.3
BIC = 471205; SABIC = 470156.4
```

```r
# print(items.gpcm)
knitr::kable(round(items.gpcm, 2), caption = "Generalized Partial Credit Model (GPCM) - Item
```

```r
# OPTION CHARACTERISTIC CURVES
plot(results.gpcm, type = "trace", which.items = c(1, 2, 3, 4),
    main = "", par.settings = simpleTheme(lty = 1:4, lwd = 2),
    auto.key = list(points = FALSE, lines = TRUE, columns = 4))  # option curves
```



Item parameters.

Item plots.

Model fit.

6.1.4 Rating Scale Model (RSM)

```r
model.rsm <- "RSM = 1-4"
results.rsm <- mirt(data = items, model = model.rsm, itemtype = "rsm",
    verbose = FALSE)
coef.rsm <- coef(results.rsm, simplify = TRUE)
items.rsm <- as.data.frame(coef.rsm$items)
print(summary(results.rsm))
```

```
      RSM    h2
```

Table 6.3: Generalized Partial Credit Model (GPCM) - Item Parameters

	a	b1	b2	b3	b4
Q7	0.89	0.36	0.02	0.67	1.01
Q10	1.06	1.17	0.59	1.36	1.69
Q11	1.51	-0.19	0.16	0.62	1.35
Q16	1.13	0.42	0.51	1.06	1.53
Q17	0.43	8.66	-0.52	0.82	0.34
Q18	1.55	-0.32	-0.27	0.59	1.14
Q19	1.44	0.18	0.19	0.82	1.49
Q20	1.03	0.93	0.30	0.89	1.15
Q21	0.97	1.14	0.31	0.84	0.68
Q22	0.97	-0.28	-0.61	-0.32	0.29
Q23	1.05	0.71	0.56	1.51	2.09
Q24	1.15	0.96	0.42	1.40	1.86
Q25	1.57	-0.95	-0.64	0.16	0.95
Q26	1.04	1.34	0.66	0.93	1.46
Q27	0.47	3.54	0.04	1.09	0.93
Q28	1.31	-0.02	0.42	0.94	1.69
Q29	1.01	0.14	-0.01	0.80	1.49
Q30	1.24	0.68	0.34	1.10	1.76
Q31	1.01	0.51	0.56	1.39	1.80
Q32	1.27	-0.36	-0.45	0.60	1.36
Q34	0.91	0.19	-0.03	0.87	1.51
Q35	1.20	-0.62	-0.63	0.24	0.75
Q36	1.44	-0.53	-0.20	0.52	1.25
Q37	1.34	-0.57	-0.38	0.42	0.96
Q38	1.75	-0.57	-0.35	0.42	0.87
Q39	1.09	-0.16	-0.59	0.24	0.58
Q40	1.39	0.39	0.20	0.94	1.11
Q41	0.40	9.15	-0.05	1.39	-0.08
Q42	1.59	-0.61	0.03	0.79	1.49
Q43	1.51	0.25	0.50	1.23	1.80
Q44	1.73	-0.24	-0.08	0.61	1.34
Q45	1.09	-0.11	-0.14	0.49	0.84
Q46	0.35	10.37	0.96	4.11	2.47
Q47	0.98	1.55	1.20	2.40	2.18
Q49	1.35	0.24	0.48	1.07	1.71
Q50	0.52	3.26	-0.90	0.64	1.01
Q51	0.77	0.70	0.27	0.74	1.37
Q52	1.60	0.38	0.48	0.95	1.67
Q53	1.24	0.22	0.14	0.88	1.32
Q54	1.31	-0.42	-0.41	0.45	0.90
Q55	0.33	3.94	1.89	3.39	3.68
Q56	1.97	-0.40	-0.13	0.50	1.23
Q57	1.25	0.61	0.29	0.93	1.44
Q58	1.24	-0.05	-0.11	0.84	1.32
Q59	1.62	0.12	0.38	1.03	1.77
Q60	0.72	2.94	0.85	1.39	2.28
Q61	1.06	0.05	-0.26	0.42	0.85
Q62	0.99	0.55	0.13	1.29	1.89
Q63	1.64	-0.13	0.29	0.99	1.60
Q64	1.19	0.11	0.38	1.04	1.50
Q65	0.93	1.28	0.70	1.30	1.76

```
Q7  0.526 0.277
Q10 0.526 0.277
Q11 0.526 0.277
Q16 0.526 0.277
Q17 0.526 0.277
Q18 0.526 0.277
Q19 0.526 0.277
Q20 0.526 0.277
Q21 0.526 0.277
Q22 0.526 0.277
Q23 0.526 0.277
Q24 0.526 0.277
Q25 0.526 0.277
Q26 0.526 0.277
Q27 0.526 0.277
Q28 0.526 0.277
Q29 0.526 0.277
Q30 0.526 0.277
Q31 0.526 0.277
Q32 0.526 0.277
Q34 0.526 0.277
Q35 0.526 0.277
Q36 0.526 0.277
Q37 0.526 0.277
Q38 0.526 0.277
Q39 0.526 0.277
Q40 0.526 0.277
Q41 0.526 0.277
Q42 0.526 0.277
Q43 0.526 0.277
Q44 0.526 0.277
Q45 0.526 0.277
Q46 0.526 0.277
Q47 0.526 0.277
Q49 0.526 0.277
Q50 0.526 0.277
Q51 0.526 0.277
Q52 0.526 0.277
Q53 0.526 0.277
Q54 0.526 0.277
Q55 0.526 0.277
Q56 0.526 0.277
Q57 0.526 0.277
Q58 0.526 0.277
Q59 0.526 0.277
Q60 0.526 0.277
```

```
Q61 0.526 0.277
Q62 0.526 0.277
Q63 0.526 0.277
Q64 0.526 0.277
Q65 0.526 0.277
Q66 0.526 0.277
Q68 0.526 0.277
Q69 0.526 0.277
Q70 0.526 0.277
Q71 0.526 0.277
Q72 0.526 0.277
Q73 0.526 0.277
Q74 0.526 0.277
Q76 0.526 0.277
Q77 0.526 0.277
Q78 0.526 0.277
Q79 0.526 0.277
Q80 0.526 0.277
Q81 0.526 0.277
Q82 0.526 0.277

SS loadings:  18.287
Proportion Var:  0.277

Factor correlations:

    RSM
RSM   1
$rotF
         RSM
Q7  0.526374
Q10 0.526374
Q11 0.526374
Q16 0.526374
Q17 0.526374
Q18 0.526374
Q19 0.526374
Q20 0.526374
Q21 0.526374
Q22 0.526374
Q23 0.526374
Q24 0.526374
Q25 0.526374
Q26 0.526374
Q27 0.526374
Q28 0.526374
```

```
Q29 0.526374
Q30 0.526374
Q31 0.526374
Q32 0.526374
Q34 0.526374
Q35 0.526374
Q36 0.526374
Q37 0.526374
Q38 0.526374
Q39 0.526374
Q40 0.526374
Q41 0.526374
Q42 0.526374
Q43 0.526374
Q44 0.526374
Q45 0.526374
Q46 0.526374
Q47 0.526374
Q49 0.526374
Q50 0.526374
Q51 0.526374
Q52 0.526374
Q53 0.526374
Q54 0.526374
Q55 0.526374
Q56 0.526374
Q57 0.526374
Q58 0.526374
Q59 0.526374
Q60 0.526374
Q61 0.526374
Q62 0.526374
Q63 0.526374
Q64 0.526374
Q65 0.526374
Q66 0.526374
Q68 0.526374
Q69 0.526374
Q70 0.526374
Q71 0.526374
Q72 0.526374
Q73 0.526374
Q74 0.526374
Q76 0.526374
Q77 0.526374
Q78 0.526374
```

```
Q79 0.526374
Q80 0.526374
Q81 0.526374
Q82 0.526374

$h2
           h2
Q7  0.2770696
Q10 0.2770696
Q11 0.2770696
Q16 0.2770696
Q17 0.2770696
Q18 0.2770696
Q19 0.2770696
Q20 0.2770696
Q21 0.2770696
Q22 0.2770696
Q23 0.2770696
Q24 0.2770696
Q25 0.2770696
Q26 0.2770696
Q27 0.2770696
Q28 0.2770696
Q29 0.2770696
Q30 0.2770696
Q31 0.2770696
Q32 0.2770696
Q34 0.2770696
Q35 0.2770696
Q36 0.2770696
Q37 0.2770696
Q38 0.2770696
Q39 0.2770696
Q40 0.2770696
Q41 0.2770696
Q42 0.2770696
Q43 0.2770696
Q44 0.2770696
Q45 0.2770696
Q46 0.2770696
Q47 0.2770696
Q49 0.2770696
Q50 0.2770696
Q51 0.2770696
Q52 0.2770696
Q53 0.2770696
```

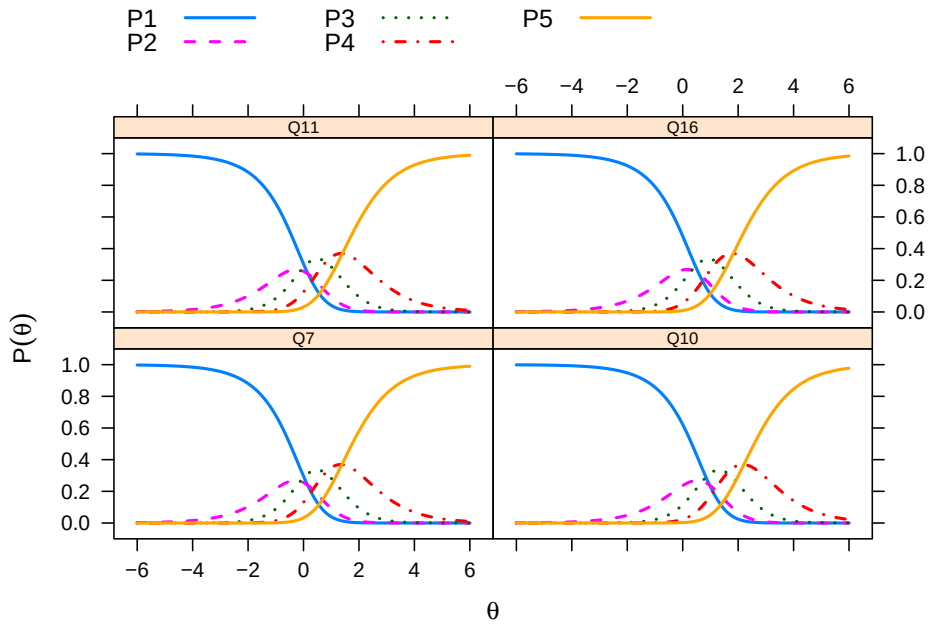
```
Q54 0.2770696
Q55 0.2770696
Q56 0.2770696
Q57 0.2770696
Q58 0.2770696
Q59 0.2770696
Q60 0.2770696
Q61 0.2770696
Q62 0.2770696
Q63 0.2770696
Q64 0.2770696
Q65 0.2770696
Q66 0.2770696
Q68 0.2770696
Q69 0.2770696
Q70 0.2770696
Q71 0.2770696
Q72 0.2770696
Q73 0.2770696
Q74 0.2770696
Q76 0.2770696
Q77 0.2770696
Q78 0.2770696
Q79 0.2770696
Q80 0.2770696
Q81 0.2770696
Q82 0.2770696

$fcor
    RSM
RSM   1
```

```r
# print(items.rsm)
knitr::kable(round(items.rsm, 2), caption = "Rating Scale Model (RSM) - Item Parameters")
```

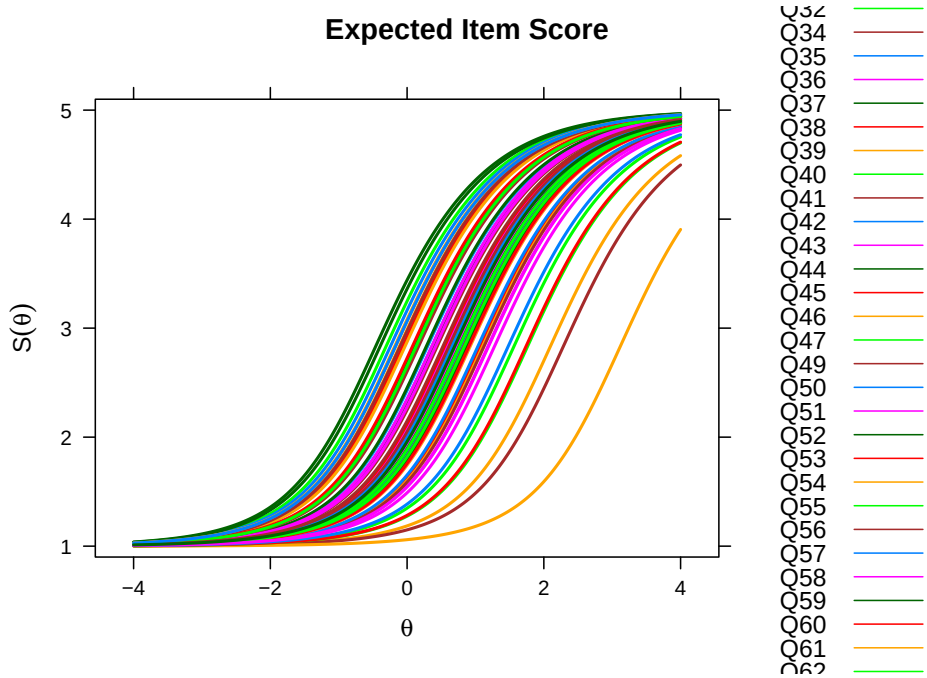
```r
plot(results.rsm, type = "trace", which.items = c(1, 2, 3, 4),
    main = "", par.settings = simpleTheme(lty = 1:4, lwd = 2),
    auto.key = list(points = FALSE, lines = TRUE, columns = 4))  # option curves
```

Table 6.4: Rating Scale Model (RSM) - Item Parameters

	a1	b1	b2	b3	b4	c
Q7	1	0.15	-0.11	0.79	1.39	0.00
Q10	1	0.15	-0.11	0.79	1.39	-0.82
Q11	1	0.15	-0.11	0.79	1.39	-0.02
Q16	1	0.15	-0.11	0.79	1.39	-0.44
Q17	1	0.15	-0.11	0.79	1.39	-1.73
Q18	1	0.15	-0.11	0.79	1.39	0.22
Q19	1	0.15	-0.11	0.79	1.39	-0.23
Q20	1	0.15	-0.11	0.79	1.39	-0.41
Q21	1	0.15	-0.11	0.79	1.39	-0.36
Q22	1	0.15	-0.11	0.79	1.39	0.82
Q23	1	0.15	-0.11	0.79	1.39	-0.70
Q24	1	0.15	-0.11	0.79	1.39	-0.74
Q25	1	0.15	-0.11	0.79	1.39	0.67
Q26	1	0.15	-0.11	0.79	1.39	-0.76
Q27	1	0.15	-0.11	0.79	1.39	-0.67
Q28	1	0.15	-0.11	0.79	1.39	-0.29
Q29	1	0.15	-0.11	0.79	1.39	-0.07
Q30	1	0.15	-0.11	0.79	1.39	-0.54
Q31	1	0.15	-0.11	0.79	1.39	-0.56
Q32	1	0.15	-0.11	0.79	1.39	0.26
Q34	1	0.15	-0.11	0.79	1.39	-0.08
Q35	1	0.15	-0.11	0.79	1.39	0.61
Q36	1	0.15	-0.11	0.79	1.39	0.25
Q37	1	0.15	-0.11	0.79	1.39	0.42
Q38	1	0.15	-0.11	0.79	1.39	0.43
Q39	1	0.15	-0.11	0.79	1.39	0.53
Q40	1	0.15	-0.11	0.79	1.39	-0.27
Q41	1	0.15	-0.11	0.79	1.39	-1.92
Q42	1	0.15	-0.11	0.79	1.39	0.06
Q43	1	0.15	-0.11	0.79	1.39	-0.54
Q44	1	0.15	-0.11	0.79	1.39	0.08
Q45	1	0.15	-0.11	0.79	1.39	0.23
Q46	1	0.15	-0.11	0.79	1.39	-2.76
Q47	1	0.15	-0.11	0.79	1.39	-1.40
Q49	1	0.15	-0.11	0.79	1.39	-0.45
Q50	1	0.15	-0.11	0.79	1.39	-0.33
Q51	1	0.15	-0.11	0.79	1.39	-0.21
Q52	1	0.15	-0.11	0.79	1.39	-0.50
Q53	1	0.15	-0.11	0.79	1.39	-0.18
Q54	1	0.15	-0.11	0.79	1.39	0.39
Q55	1	0.15	-0.11	0.79	1.39	-1.20
Q56	1	0.15	-0.11	0.79	1.39	0.19
Q57	1	0.15	-0.11	0.79	1.39	-0.41
Q58	1	0.15	-0.11	0.79	1.39	0.00
Q59	1	0.15	-0.11	0.79	1.39	-0.40
Q60	1	0.15	-0.11	0.79	1.39	-1.38
Q61	1	0.15	-0.11	0.79	1.39	0.25
Q62	1	0.15	-0.11	0.79	1.39	-0.40
Q63	1	0.15	-0.11	0.79	1.39	-0.25
Q64	1	0.15	-0.11	0.79	1.39	-0.30
Q65	1	0.15	-0.11	0.79	1.39	-0.83




```
plot(results.rsm, type = "itemscore", theta_lim = c(-4, 4), lwd = 2,
    facet_items = FALSE)  # item scoring traceline
```



```
# plot(results.rsm, type = 'infotrace', which.items =
# c(1,2,3,4), main = '', par.settings = simpleTheme(lwd=2))
# # item information functions (IIF) plot(results.rsm, type
# = 'info', theta_lim = c(-4,4), lwd=2) # test information
# curve plot(results.rsm, type = 'SE', theta_lim = c(-4,4),
# lwd=2) # test standard errors
```

6.1.5 Graded Response Model (GRM)

```
model.grm <- "GRM = 1-4"
results.grm <- mirt(data = items, model = model.grm, itemtype = "graded",
    verbose = FALSE)
coef.grm <- coef(results.grm, simplify = TRUE)
items.grm <- as.data.frame(coef.grm$items)
print(summary(results.grm))
```

```
      GRM    h2
Q7  0.764 0.584
Q10 0.736 0.542
Q11 0.843 0.711
Q16 0.691 0.478
Q17 0.000 0.000
Q18 0.000 0.000
Q19 0.000 0.000
Q20 0.000 0.000
Q21 0.000 0.000
Q22 0.000 0.000
Q23 0.000 0.000
Q24 0.000 0.000
Q25 0.000 0.000
Q26 0.000 0.000
Q27 0.000 0.000
Q28 0.000 0.000
Q29 0.000 0.000
Q30 0.000 0.000
Q31 0.000 0.000
Q32 0.000 0.000
Q34 0.000 0.000
Q35 0.000 0.000
Q36 0.000 0.000
Q37 0.000 0.000
Q38 0.000 0.000
Q39 0.000 0.000
Q40 0.000 0.000
Q41 0.000 0.000
```

```
Q42 0.000 0.000
Q43 0.000 0.000
Q44 0.000 0.000
Q45 0.000 0.000
Q46 0.000 0.000
Q47 0.000 0.000
Q49 0.000 0.000
Q50 0.000 0.000
Q51 0.000 0.000
Q52 0.000 0.000
Q53 0.000 0.000
Q54 0.000 0.000
Q55 0.000 0.000
Q56 0.000 0.000
Q57 0.000 0.000
Q58 0.000 0.000
Q59 0.000 0.000
Q60 0.000 0.000
Q61 0.000 0.000
Q62 0.000 0.000
Q63 0.000 0.000
Q64 0.000 0.000
Q65 0.000 0.000
Q66 0.000 0.000
Q68 0.000 0.000
Q69 0.000 0.000
Q70 0.000 0.000
Q71 0.000 0.000
Q72 0.000 0.000
Q73 0.000 0.000
Q74 0.000 0.000
Q76 0.000 0.000
Q77 0.000 0.000
Q78 0.000 0.000
Q79 0.000 0.000
Q80 0.000 0.000
Q81 0.000 0.000
Q82 0.000 0.000

SS loadings:  2.316
Proportion Var:  0.035

Factor correlations:

    GRM
GRM   1
```

```
$rotF
          GRM
Q7  0.7644889
Q10 0.7362822
Q11 0.8434104
Q16 0.6913146
Q17 0.0000000
Q18 0.0000000
Q19 0.0000000
Q20 0.0000000
Q21 0.0000000
Q22 0.0000000
Q23 0.0000000
Q24 0.0000000
Q25 0.0000000
Q26 0.0000000
Q27 0.0000000
Q28 0.0000000
Q29 0.0000000
Q30 0.0000000
Q31 0.0000000
Q32 0.0000000
Q34 0.0000000
Q35 0.0000000
Q36 0.0000000
Q37 0.0000000
Q38 0.0000000
Q39 0.0000000
Q40 0.0000000
Q41 0.0000000
Q42 0.0000000
Q43 0.0000000
Q44 0.0000000
Q45 0.0000000
Q46 0.0000000
Q47 0.0000000
Q49 0.0000000
Q50 0.0000000
Q51 0.0000000
Q52 0.0000000
Q53 0.0000000
Q54 0.0000000
Q55 0.0000000
Q56 0.0000000
Q57 0.0000000
Q58 0.0000000
```

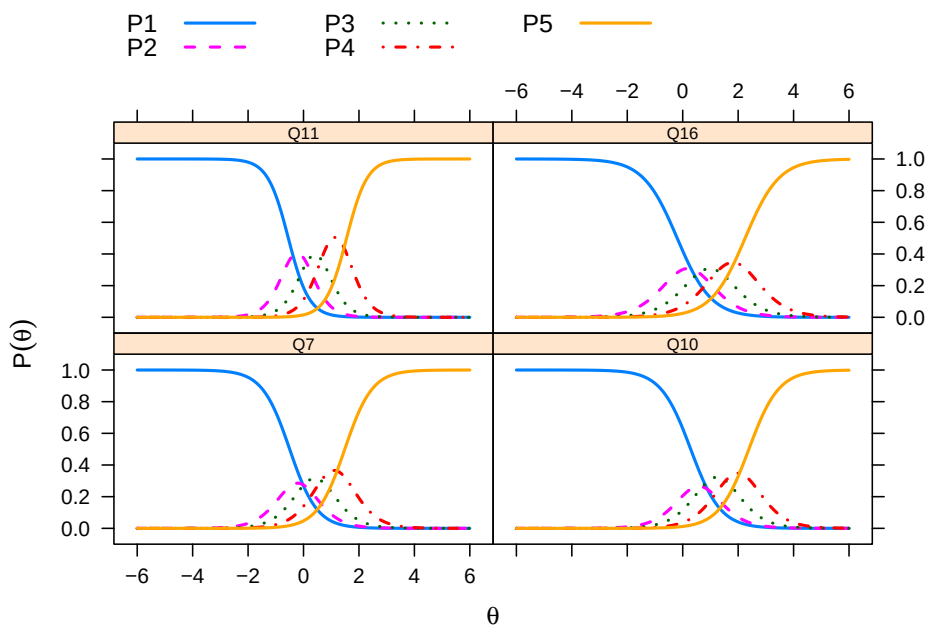
```
Q59 0.0000000
Q60 0.0000000
Q61 0.0000000
Q62 0.0000000
Q63 0.0000000
Q64 0.0000000
Q65 0.0000000
Q66 0.0000000
Q68 0.0000000
Q69 0.0000000
Q70 0.0000000
Q71 0.0000000
Q72 0.0000000
Q73 0.0000000
Q74 0.0000000
Q76 0.0000000
Q77 0.0000000
Q78 0.0000000
Q79 0.0000000
Q80 0.0000000
Q81 0.0000000
Q82 0.0000000

$h2
           h2
Q7  0.5844433
Q10 0.5421115
Q11 0.7113411
Q16 0.4779159
Q17 0.0000000
Q18 0.0000000
Q19 0.0000000
Q20 0.0000000
Q21 0.0000000
Q22 0.0000000
Q23 0.0000000
Q24 0.0000000
Q25 0.0000000
Q26 0.0000000
Q27 0.0000000
Q28 0.0000000
Q29 0.0000000
Q30 0.0000000
Q31 0.0000000
Q32 0.0000000
Q34 0.0000000
```

```
Q35 0.0000000
Q36 0.0000000
Q37 0.0000000
Q38 0.0000000
Q39 0.0000000
Q40 0.0000000
Q41 0.0000000
Q42 0.0000000
Q43 0.0000000
Q44 0.0000000
Q45 0.0000000
Q46 0.0000000
Q47 0.0000000
Q49 0.0000000
Q50 0.0000000
Q51 0.0000000
Q52 0.0000000
Q53 0.0000000
Q54 0.0000000
Q55 0.0000000
Q56 0.0000000
Q57 0.0000000
Q58 0.0000000
Q59 0.0000000
Q60 0.0000000
Q61 0.0000000
Q62 0.0000000
Q63 0.0000000
Q64 0.0000000
Q65 0.0000000
Q66 0.0000000
Q68 0.0000000
Q69 0.0000000
Q70 0.0000000
Q71 0.0000000
Q72 0.0000000
Q73 0.0000000
Q74 0.0000000
Q76 0.0000000
Q77 0.0000000
Q78 0.0000000
Q79 0.0000000
Q80 0.0000000
Q81 0.0000000
Q82 0.0000000
```

```
$fcor
    GRM
GRM   1
```

```r
# print(items.grm)
knitr::kable(round(items.grm, 2), caption = "Graded Response Model (GRM) - Item Parameters")
```

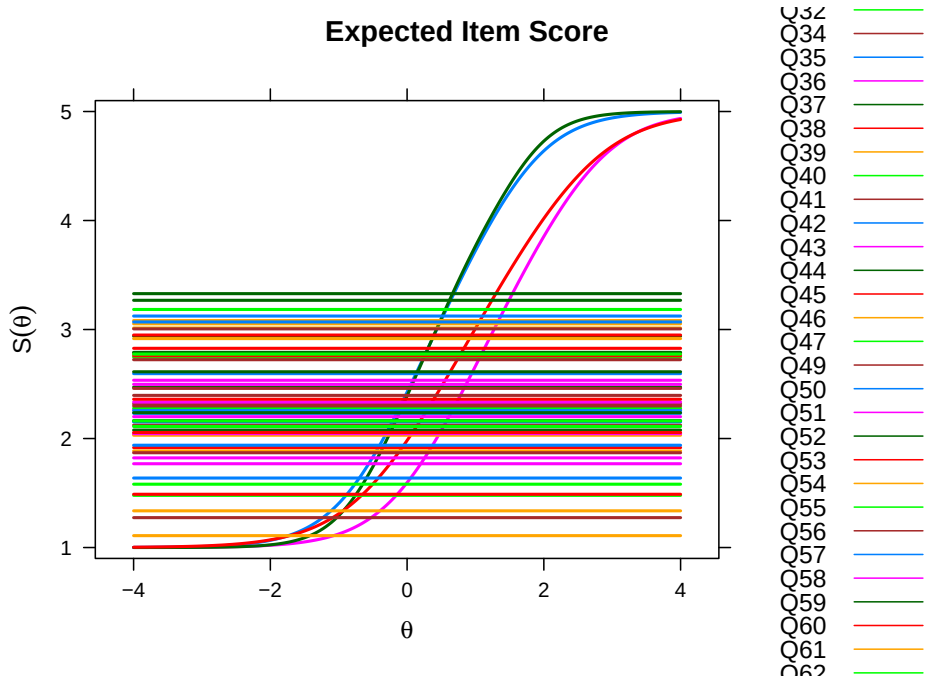
```r
plot(results.grm, type = "trace", which.items = c(1, 2, 3, 4),
    main = "", par.settings = simpleTheme(lty = 1:4, lwd = 2),
    auto.key = list(points = FALSE, lines = TRUE, columns = 4))  # option curves
```




```r
plot(results.grm, type = "itemscore", theta_lim = c(-4, 4), lwd = 2,
    facet_items = FALSE)  # item scoring traceline
```

Table 6.5: Graded Response Model (GRM) - Item Parameters

	a1	d1	d2	d3	d4
Q7	2.02	1.01	-0.17	-1.47	-3.01
Q10	1.85	-0.53	-1.62	-2.96	-4.42
Q11	2.67	1.46	-0.26	-1.93	-4.15
Q16	1.63	0.32	-0.95	-2.22	-3.67
Q17	0.00	-1.95	-2.17	-2.58	-3.19
Q18	0.00	1.00	0.26	-0.75	-1.91
Q19	0.00	0.41	-0.34	-1.24	-2.49
Q20	0.00	0.01	-0.61	-1.34	-2.26
Q21	0.00	-0.04	-0.60	-1.22	-1.92
Q22	0.00	1.41	0.80	0.15	-0.85
Q23	0.00	-0.04	-0.93	-2.07	-3.43
Q24	0.00	-0.23	-0.93	-1.98	-3.21
Q25	0.00	1.71	0.85	-0.19	-1.48
Q26	0.00	-0.42	-1.05	-1.71	-2.71
Q27	0.00	-0.43	-0.90	-1.57	-2.40
Q28	0.00	0.51	-0.47	-1.43	-2.76
Q29	0.00	0.66	-0.13	-1.08	-2.31
Q30	0.00	0.00	-0.70	-1.65	-2.92
Q31	0.00	0.15	-0.79	-1.85	-3.05
Q32	0.00	1.14	0.38	-0.74	-2.08
Q34	0.00	0.67	-0.14	-1.12	-2.34
Q35	0.00	1.46	0.70	-0.26	-1.34
Q36	0.00	1.16	0.28	-0.72	-2.00
Q37	0.00	1.29	0.46	-0.50	-1.63
Q38	0.00	1.24	0.47	-0.49	-1.53
Q39	0.00	1.17	0.55	-0.28	-1.24
Q40	0.00	0.25	-0.42	-1.27	-2.18
Q41	0.00	-2.13	-2.41	-2.87	-3.41
Q42	0.00	1.11	0.04	-1.08	-2.44
Q43	0.00	0.18	-0.74	-1.83	-3.15
Q44	0.00	0.83	0.08	-0.88	-2.19
Q45	0.00	0.91	0.16	-0.65	-1.62
Q46	0.00	-2.74	-3.37	-4.50	-5.49
Q47	0.00	-0.86	-1.89	-3.25	-4.38
Q49	0.00	0.25	-0.65	-1.62	-2.90
Q50	0.00	-0.09	-0.44	-1.12	-2.05
Q51	0.00	0.38	-0.37	-1.16	-2.25
Q52	0.00	0.07	-0.70	-1.56	-2.86
Q53	0.00	0.46	-0.30	-1.21	-2.30
Q54	0.00	1.19	0.44	-0.52	-1.59
Q55	0.00	-0.64	-1.63	-2.80	-4.12
Q56	0.00	0.98	0.21	-0.72	-2.02
Q57	0.00	0.09	-0.58	-1.41	-2.51
Q58	0.00	0.76	-0.01	-1.05	-2.21
Q59	0.00	0.32	-0.55	-1.58	-2.99
Q60	0.00	-1.14	-1.75	-2.53	-3.73
Q61	0.00	0.87	0.21	-0.59	-1.59
Q62	0.00	0.28	-0.49	-1.62	-2.95
Q63	0.00	0.58	-0.38	-1.44	-2.75
Q64	0.00	0.46	-0.47	-1.45	-2.59
Q65	0.00	-0.36	-1.12	-1.98	-3.09




```
# plot(results.rsm, type = 'infotrace', which.items =
# c(1,2,3,4), main = '', par.settings = simpleTheme(lwd=2))
# # item information functions (IIF) plot(results.rsm, type
# = 'info', theta_lim = c(-4,4), lwd=2) # test information
# curve plot(results.rsm, type = 'SE', theta_lim = c(-4,4),
# lwd=2) # test standard errors
```

In the Graded Response Model, between category threshold parameters are ordered.

6.1.6 Trait scores

```
# attach factor scores pcmscores <- fscores(results.pcm)
# gpcmscores <- fscores(results.gpcm) rsmscores <-
# fscores(results.rsm) grmscores <- fscores(results.grm)
# pew <- cbind(pew, pcmscores) pew <- cbind(pew,
# gpcmscores) pew <- cbind(pew, rsmscores) pew <-
# cbind(pew, grmscores) head(items) scores <- pew[ ,5:9]
# corrs2 <- cor(scores, method = c('pearson'))
# knitr::kable(round(corrs2,2), caption = 'Correlation
# Matrix')
```

6.2 Summary

Chapter 7

Reliability

In this chapter, we addresses the reliability of the SHAS based on the 66 piloted items. We begin by defining reliability and discussing its importance. Then we report the results of our analyses of the reliability of the piloted SHAS items from both classical test theory (CTT) and item response theory (IRT) frameworks. Since these are cross-sectional data taken at one point in time, our work is limited to *internal consistency* reliability which focuses on the relationships between the items at one point in time rather than relationships between items or scores between two or more time points. We conclude by considering what this information contributes to improved SHAS instrumentation.

7.1 About reliability

Reliability is the extent to which items and their scores are consistently measuring the trait of interest. It is evidence that two examinees who provide the same responses to the same items, and by extension receive the same total scores, truly have the same level or amount of, or exposure to, the trait.

7.2 Reliability in Classical Test Theory (CTT)

One way to think about reliability in classical test theory is the ratio of signal to noise in the observed data. Do the various item responses and scores provide a true sort of examinees on the trait, or is all this variation merely noise?

The foundation of internal consistency reliability in CTT is relationships between item responses: correlations. Strong correlations among the items are evidence that the items are measuring the same construct. Thus we begin by examining correlation matrices.

```r
# HEATMAP OF CORRELATIONS
psych::corPlot(ext_items_l, upper = FALSE)
```

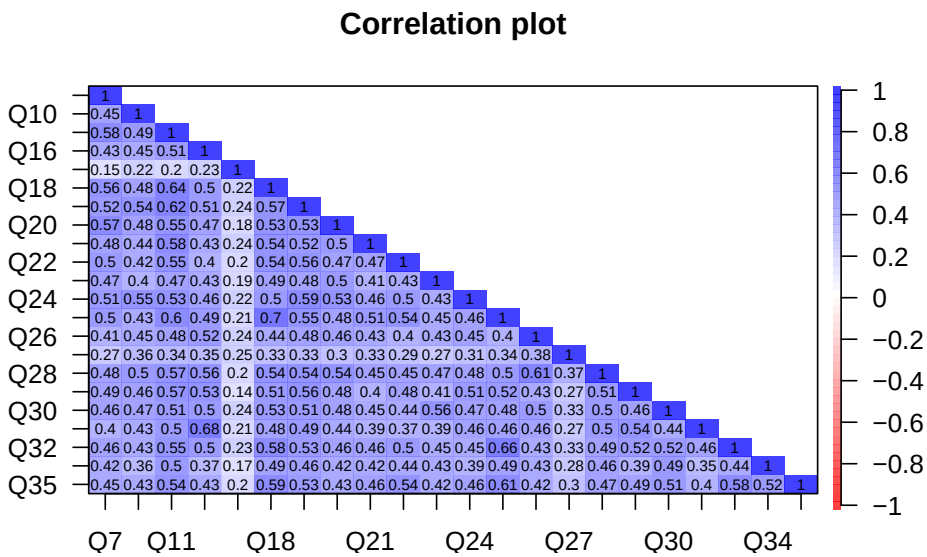


Figure 7.1: Correlation Matrix of External Events Items

Figure 7.1 is a correlation matrix of the External Events items for the *3,174* examinees who responded to all of these items. Each cell reports the correlation coefficient between scores of the two items. All of the cells are color coded to the scale of correlations which range from 0 (indicating no relationship) to 1 (indicating a perfect linear relationship). The cells along the diagonal are all 1.0 because they represent the correlation of the item with itself. Only one half of the matrix is necessary to report all the correlations among all the items.

The plot offers several interesting pieces of information. The first is that Q17–"Being treated as 'less than' because of my sexual orientation–does not quite fit. Responses to this item correlate weakly with responses to all the other items. It is possible that so few examinees experience this event that the responses vary very little, which would attenuate any correlation. Or it may be measuring something different than what the other items in this set. Similar could be said of Q27–"Being denied opportunities to serve/volunteer because of my gender."

```r
# HEATMAP OF CORRELATIONS
psych::corPlot(int_items_l, upper = FALSE)
```

Figure 7.2 is a correlation matrix of the Internal States items for the *3,051* examinees who responded to all of these items. This plot is interesting in that most of the correlations are in the moderate (0.50) range. They are sorting examinees in a similar way, but together they do not measure exactly the same

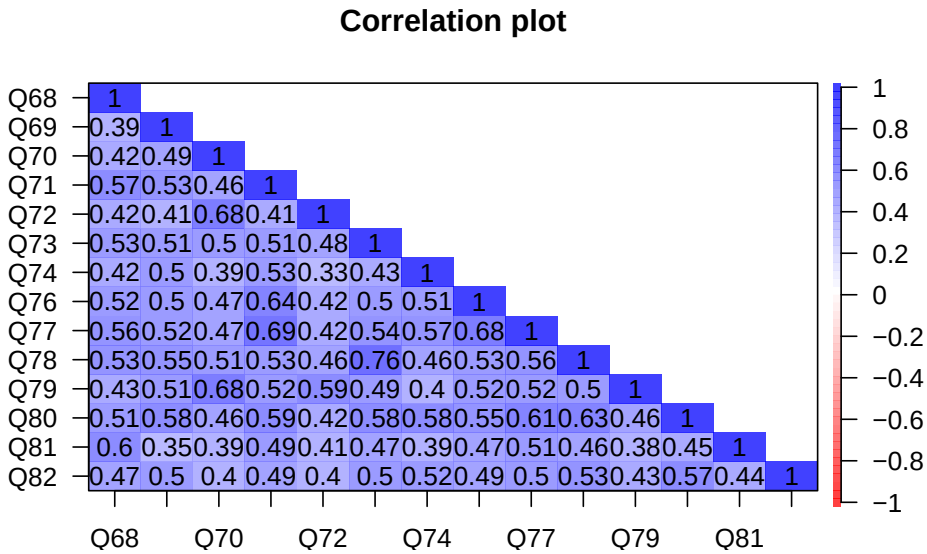


Figure 7.2: Correlation Matrix of Internal States Items

thing.

```
# HEATMAP OF CORRELATIONS
psych::corPlot(items_l, upper = FALSE)
```

This final matrix, Figure 7.3, reports the correlations among all of the Likert items among the *2,979* examinees who responded to all 66 items. Note that missing data are a challenge here, as these plots require complete data.

This plot offers several interesting insights. First, the preponderance of blue indicates that everything is positively correlated. This makes some sense because the items were worded and all the response categories were coded in the same direction.

Second, in this plot there are too many items and correlation coefficients in one place to be distinguishable, but with the colors it is possible to see an overall pattern. This plot orders the items by their order on the questionnaire so we can see how the External Events items correlate with the Internal Events items: moderately. It is possible that these are not two different scales.

Finally, we see at a glance a handful of items that correlate very weakly with all the other items. These items should be scrutinized for refinement and possible omission from the analysis.

Cronbach's coefficient alpha. Arguably the most well-known statistic for estimating internal consistency reliability is Cronbach's coefficient alpha (citation). This is a measure of the average inter-item correlation (?). It is also a measure of the ratio of signal to noise (X = T + e). A ballpark cutoff value is

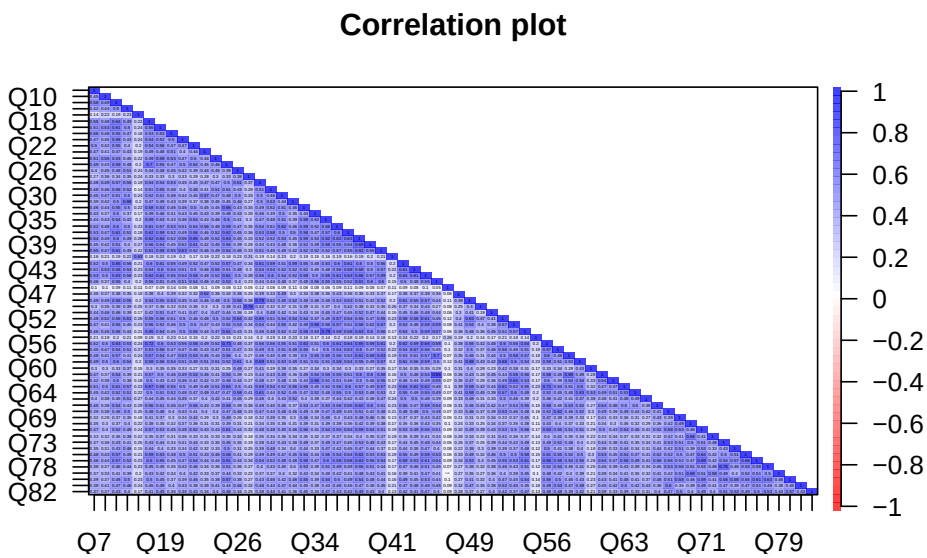


Figure 7.3: Correlation Matrix of All 66 Likert SHAS Items

0.70 [Nunnally and Bernstein, 1994].

```
# CRONBACH'S ALPHA
reliability(items)  # , itemal = TRUE, NA.Delete = TRUE, ml = TRUE)
```

```
Number of Items
66

Number of Examinees
2979

Coefficient Alpha
0.981
```

This output is not pretty, but it does report a Cronbach's alpha of 0.98. This is a high level of internal consistency reliability.

7.3 Reliability in Item Response Theory (IRT)

Wright and Masters (1982) developed reliability measures based on the Rasch measurement model, which have a different concept from classical test theory (e.g., Cronbach's alpha). Reliability is estimated both for persons and items, by means of Person Separation Reliability (PSR). This is an estimate of how well this instrument can distinguish respondents on the measured variable. In parallel, Item Separation Reliability (ISR) is an indication of how well items are separated by the persons taking the test (Wright & Stone, 1999). The cutoff of

separation reliabilities is > 0.80 (Bond & Fox, 2015).

7.4 Summary

Chapter 8

Dimensionality

This chapter deals with the topic of dimensionality. It begins with a definition of dimensionality and explains how it is important. It outlines how to investigate it and suggests what counts as good evidence. Then it reports our study of the dimensionality of the SHAS.

8.1 About Dimensionality

Dimensionality is the study of the dimensions (also known as "factors", "components", and "latent traits") that the items are designed and assembled to collectively measure. This study tends to take two forms. One is exploration of what dimensions a set of items apparently measure, regardless of what they intended to measure. The other is what evidence exist to confirm that a set of items actually measure the dimension they are intended to measure.

Dimensionality is important for the reporting of scores. Scores, as summary measures of a set of items, are only meaningful to the extent that they enjoy evidence that they measure the dimension(s) they are designed to measure. Scores, in turn, are important to the extent that they support decisions that have consequences in the real world. Scores are needed to decide who gets to practice medicine, or fly an airplane.

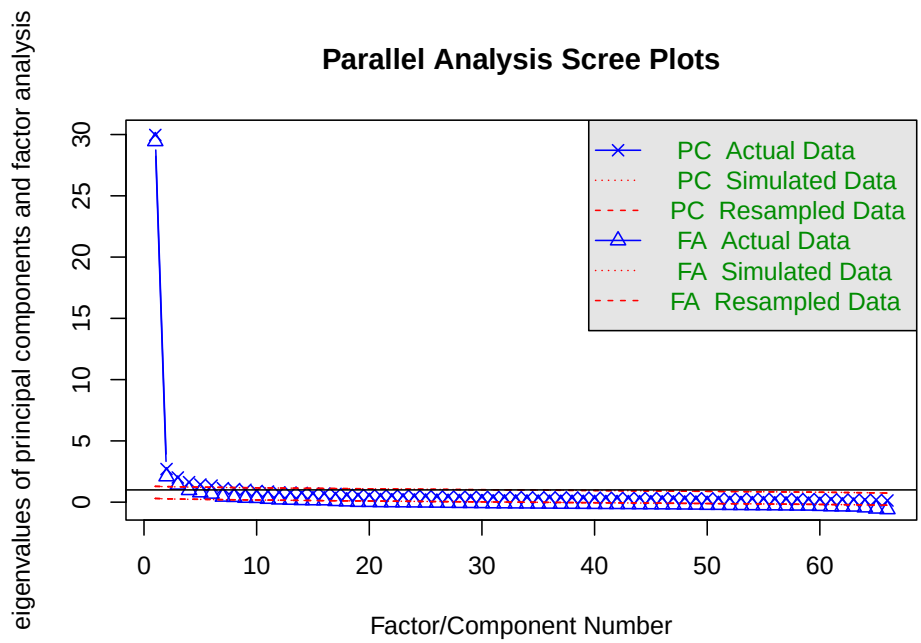
Dimensionality is an important part of the validation process for psychological measures [Embretson and Reise, 2000].

The methodology commonly used to investigate dimensionality is factor analysis. To assess the dimensionality of these 66 piloted SHAS items we use exploratory factor analysis and confirmatory factor analysis (CFA) and PCAR.

8.2 Exploratory Factor Analysis (EFA)

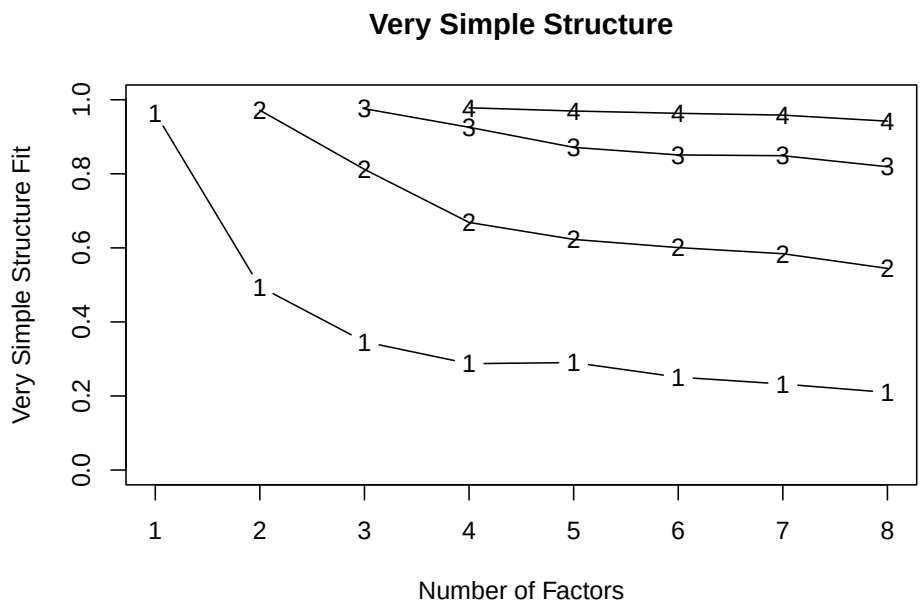
```
# LOAD PACKAGES
library(lavaan)
library(psych)
library(psychTools)

# LOAD DATA all <- read.csv('data/all.csv')
items <- read.csv("data/items.csv")
ext_items <- read.csv("data/ext_items.csv")
int_items <- read.csv("data/int_items.csv")
```

```
# TEST FOR NUMBER OF FACTORS IN DATA USING PARALLEL
# ANALYSIS AND VERY SIMPLE STRUCTURE (VSS)
fa.parallel(items)
```




```
Parallel analysis suggests that the number of factors =  16  and the number of components =
```

```
vss(items)
```




```
Very Simple Structure
Call: vss(x = items)
VSS complexity 1 achieves a maximimum of 0.96  with  1  factors
VSS complexity 2 achieves a maximimum of 0.97  with  2  factors

The Velicer MAP achieves a minimum of 0.01  with  6  factors
BIC achieves a minimum of  -788.85  with  8  factors
Sample Size adjusted BIC achieves a minimum of  4438.02  with  8  factors

Statistics by number of factors
  vss1 vss2    map  dof chisq prob sqresid  fit RMSEA   BIC SABIC complex
1 0.96 0.00 0.0091 2079 37231    0      33 0.96 0.072 20438 27044     1.0
2 0.50 0.97 0.0068 2014 29480    0      26 0.97 0.065 13211 19611     1.6
3 0.35 0.81 0.0058 1950 24035    0      23 0.98 0.059  8283 14479     2.1
4 0.29 0.67 0.0056 1887 20947    0      20 0.98 0.056  5704 11700     2.5
5 0.29 0.62 0.0055 1825 17804    0      18 0.98 0.052  3062  8861     2.7
6 0.25 0.60 0.0052 1764 15824    0      16 0.98 0.050  1575  7180     2.8
7 0.23 0.58 0.0056 1704 13729    0      15 0.98 0.047   -35  5379     2.9
8 0.21 0.54 0.0055 1645 12499    0      14 0.98 0.045  -789  4438     3.0
  eChisq  SRMR eCRMS  eBIC
1  37967 0.052 0.053 21173
2  23187 0.041 0.042  6918
3  16097 0.034 0.036   345
4  12506 0.030 0.032 -2737
5   9591 0.026 0.029 -5151
6   7329 0.023 0.025 -6921
```

```
7   5949 0.021 0.023 -7815
8   5053 0.019 0.022 -8235
```

```r
# PRINCIPAL COMPONENTS ANALYSIS
principal(items)
```

```
Principal Components Analysis
Call: principal(r = items)
Standardized loadings (pattern matrix) based upon correlation matrix
     PC1    h2   u2 com
Q7  0.67 0.451 0.55   1
Q10 0.64 0.404 0.60   1
Q11 0.77 0.598 0.40   1
Q16 0.69 0.481 0.52   1
Q17 0.33 0.112 0.89   1
Q18 0.78 0.604 0.40   1
Q19 0.75 0.569 0.43   1
Q20 0.68 0.466 0.53   1
Q21 0.68 0.467 0.53   1
Q22 0.70 0.488 0.51   1
Q23 0.64 0.406 0.59   1
Q24 0.66 0.440 0.56   1
Q25 0.76 0.583 0.42   1
Q26 0.66 0.430 0.57   1
Q27 0.47 0.222 0.78   1
Q28 0.73 0.529 0.47   1
Q29 0.69 0.474 0.53   1
Q30 0.70 0.496 0.50   1
Q31 0.65 0.417 0.58   1
Q32 0.73 0.539 0.46   1
Q34 0.65 0.428 0.57   1
Q35 0.74 0.542 0.46   1
Q36 0.77 0.590 0.41   1
Q37 0.76 0.572 0.43   1
Q38 0.80 0.647 0.35   1
Q39 0.72 0.515 0.48   1
Q40 0.75 0.564 0.44   1
Q41 0.30 0.089 0.91   1
Q42 0.77 0.594 0.41   1
Q43 0.74 0.542 0.46   1
Q44 0.80 0.642 0.36   1
Q45 0.72 0.516 0.48   1
Q46 0.15 0.021 0.98   1
Q47 0.52 0.269 0.73   1
Q49 0.72 0.520 0.48   1
Q50 0.53 0.281 0.72   1
```

```
Q51 0.62 0.380 0.62   1
Q52 0.76 0.577 0.42   1
Q53 0.73 0.535 0.46   1
Q54 0.75 0.568 0.43   1
Q55 0.27 0.075 0.93   1
Q56 0.82 0.670 0.33   1
Q57 0.72 0.521 0.48   1
Q58 0.73 0.533 0.47   1
Q59 0.76 0.579 0.42   1
Q60 0.47 0.224 0.78   1
Q61 0.72 0.511 0.49   1
Q62 0.65 0.424 0.58   1
Q63 0.77 0.590 0.41   1
Q64 0.71 0.503 0.50   1
Q65 0.61 0.373 0.63   1
Q66 0.71 0.498 0.50   1
Q68 0.68 0.459 0.54   1
Q69 0.57 0.323 0.68   1
Q70 0.56 0.309 0.69   1
Q71 0.74 0.544 0.46   1
Q72 0.54 0.294 0.71   1
Q73 0.64 0.405 0.59   1
Q74 0.60 0.359 0.64   1
Q76 0.74 0.544 0.46   1
Q77 0.77 0.598 0.40   1
Q78 0.66 0.442 0.56   1
Q79 0.60 0.365 0.64   1
Q80 0.68 0.466 0.53   1
Q81 0.65 0.419 0.58   1
Q82 0.61 0.368 0.63   1

                 PC1
SS loadings    29.96
Proportion Var  0.45

Mean item complexity =  1
Test of the hypothesis that 1 component is sufficient.

The root mean square of the residuals (RMSR) is  0.05
 with the empirical chi square  38864.3  with prob <  0

Fit based upon off diagonal values = 0.99
```

```
# PRINCIPAL AXIS FACTORING library(psych) fit <-
# factor.pa(items, nfactors=3, rotation='varimax') fit #
# print results
```

```
# PRINCIPAL COMPONENTS ANALYSIS entering raw data and
# extracting PCs from the correlation matrix fit <-
# princomp(items, cor=TRUE) summary(fit) # print variance
# accounted for loadings(fit) # pc loadings
# plot(fit,type='lines') # scree plot fit$scores # the
# principal components biplot(fit)
```

```
# DETERMINE NUMBER OF FACTORS TO EXTRACT library(nFactors)
# ev <- eigen(cor(items)) # get eigenvalues ap <-
# parallel(subject=nrow(items),var=ncol(items),
# rep=100,cent=.05) nS <- nScree(x=ev$values,
# aparallel=ap$eigen$qevpea) plotnScree(nS)
```

```
# MAXIMUM LIKELIHOOD FACTOR ANALYSIS entering raw data and
# extracting 3 factors, with varimax rotation fit <-
# factanal(items, 3, rotation='varimax') print(fit,
# digits=2, cutoff=.3, sort=TRUE) # plot factor 1 by factor
# 2 load <- fit$loadings[,1:2] plot(load,type='n') # set up
# plot text(load,labels=names(items),cex=.7) # add variable
# names
```

8.3 Confirmatory Factor Analysis (CFA)

The purpose of confirmatory factor analysis is, as the name implies, to confirm that items measure the factor(s) or trait(s) for which the items were designed and intended to measure.

In our use of CFA we used several statistics and fit indices:

- Root Mean Square Error of Approximation (RMSEA)
- Comparative Fit Index (CFI)
- Tucker–Lewis Index (TLI), and
- Standardized Root-Mean-Square Residual (SRMR)

Based on published criteria (Hu & Bentler, 1999; Wang & Wang, 2019), we used the following standards for good fit:

- CFI > 0.95
- TLI > 0.95
- RMSEA < 0.05
- SRMR < 0.08

We use the MLM estimator [Brown, 2006].

```
# ltm::factor.scores(items) # Computation of factor scores
# for grm, ltm, rasch and tpm models
```

```
# library(lavaan) MODEL WITH ONE COMMON FACTOR

# wb_1factor <- ' #start of model

# latent variable definitions (common factors) wellbeing =~
# NA*sense.peace + NA*sense.wonder + NA*sense.gratitude +
# NA*sense.purpose

# latent variable variances wellbeing ~~ 1*wellbeing

# latent variable covariances latent variable means

# manifest variable variances (uniquenesses) sense.peace ~~
# sense.peace sense.wonder ~~ sense.wonder sense.gratitude
# ~~ sense.gratitude sense.purpose ~~ sense.purpose

# manifest variable covariances (uniquenesses)

# manifest variable means sense.peace ~ 1 sense.wonder ~ 1
# sense.gratitude ~ 1 sense.purpose ~ 1 end of model

# CFA1 <- lavaan(wb_1factor, data=items, estimator = 'MLM')
# # estimate model summary(CFA1, standardized=TRUE,
# fit.measures=TRUE)
```

The results of CFA, using MLR estimation, confirmed the second order factor model, in line with the original structure of the scale (see Olufadi, 2017), because the values of the indices were above the acceptable threshold [2 (167) = 284.032, p < 0.001; RMSEA = 0.030 (90% CI = 0.024-0.036), CFI = 0.983, TLI = 0.981, SRMR = 0.023], compared to the unidimensional model [2 (170) = 316.949, p < 0.001; RMSEA = 0.034 (90% CI = 0.028-0.039), CFI = 0.978, TLI = 0.976, SRMR = 0.023] and a 3-uncorrelated-factors model [2 (170) = 2427.628, p < 0.001; RMSEA = 0.132 (90% CI = 0.127-0.136), CFI = 0.669, TLI = 0.631, SRMR = 0.345]. Based on RMSEA, CFI, TLI, and SRMR, the results indicated that the model provided satisfactory representations of the underlying structure of the psychological well-being construct. All items loaded significantly (ranging from 0.535 to 0.817) in relation to each first order factor, at a p < 0.01 significance level. These results of the CFA offer evidence that the four items measure one dimension of well-being.

8.4 Summary

Chapter 9

Scale Analysis

In this chapter, we examine the technical quality of the scales derived from the SHAS piloted items. We begin by defining scaling and discussing the importance of scale analysis. We report the results of our scale analysis from within the item response theory (IRT) framework. We conclude by considering what this information contributes to improved SHAS instrumentation.

9.1 The Item-Person Map

The item-person map (also known as the Wright map) is a tool for visualizing the results of a calibration of items and persons. The Wright map was originally developed for use in test development but it also helps illuminate the results of surveys [Bulut, 2021]. As Bulut [2021] explains:

"The Partial Credit Model (PCM), which is an extension of the Rasch model for polytomous item responses, can be used for survey development and validation (Green & Frantom, 2002). We can analyze survey items using PCM and obtain item thresholds for each item. These thresholds indicate the magnitude of the latent trait required to select a particular response option (e.g., selecting disagree over strongly disagree). Using an item-person map, we can see thresholds for the items as well as the distribution of the latent trait for the respondents. This plot is quite useful for analyzing the alignment (or match) between the items and respondents answering the items."

```
# ITEM-PERSON MAP #1: FOR ALL THE SHAS ITEMS
source("code/itempersonmap.R")
library("mirt")
items <- read.csv("data/items.csv")
mod <- mirt(items, 1, itemtype = "Rasch", technical = list(removeEmptyRows = TRUE),
    verbose = FALSE)  # estimate the Partial Credit Model
itempersonmap(mod)  # create the item-person map
```

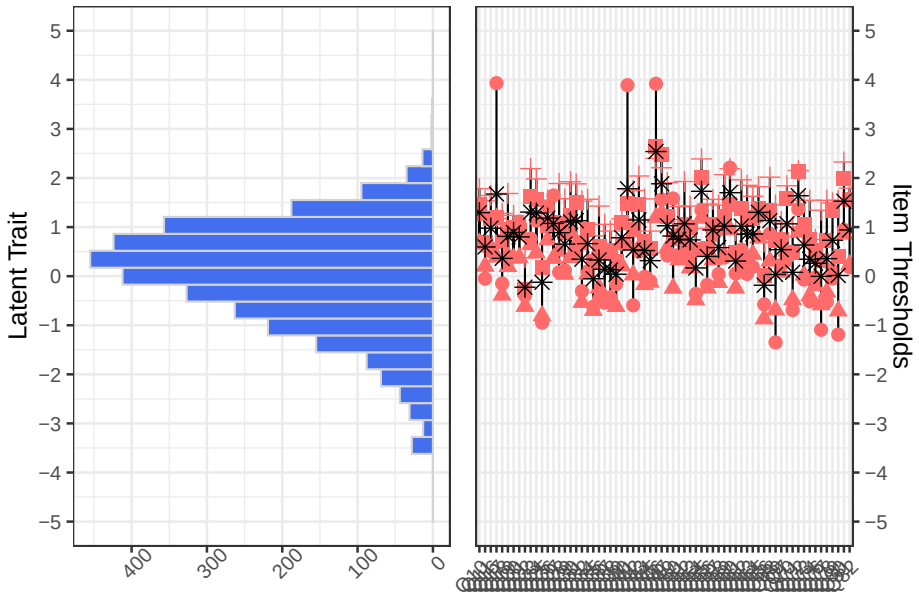


Figure 9.1: Item-Person Map: All Likert Items

Figure 9.1 is an item-person map of all 66 Likert SHAS items against all of the examinees. In the plot, red points indicate item thresholds for each item (three thresholds separating four response categories) and the asterisk indicates the average value of the thresholds for each item. The higher these thresholds, the more spiritual abuse the examinee has experienced. On the left-hand side of the plot, we also see the distribution of examinees along the latent trait (i.e., the construct of spiritual abuse).

This plot offers several insights at a general level. First, the plot of examinees on the left suggests that the latent trait of spiritual abuse is normally distributed, with a slight negative skew. Very few examinees reported very low or very high levels of exposure to spiritual abuse; most reported a moderate to slightly higher than average level of spiritual abuse. Second, with this plot it is difficult to distinguish individual items but it does show that they tend to clusteraround this "slightly higher than average" level of spiritual abuse, with none capturing a very low level of abuse and just a handful of items measuring much higher than average levels of spiritual abuse. Three items had response categories that captured very high levels of spiritual abuse.

9.2 Information curves

These provide important information about where along the scale of the latent trait the item or set of items provides the most reliable information. If there are too few "easy" or "difficult" items, then these extreme scores for examinees are less reliable.

9.3 Person Separation Reliability

9.4 Item Separation

9.5 Summary

Chapter 10

Discussion

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